

Bunching at a Moving Threshold: Firm Responses to Romania’s Micro-Enterprise Tax, 2014–2025

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Abstract

Romanian companies with revenue below a ceiling set in euro pay 1 to 3 percent of revenue instead of 16 percent of profit. The ceiling moved six times between 2013 and 2026, which gives a counterfactual density at each notch from years when no ceiling was there, and repeated reassignment of the same firms between a revenue base and a profit base. In the accounts of every company, 9.6 million firm-years, reported revenue piles up just below the ceiling, follows it when it moves, and within a fixed euro ceiling tracks its lei value at the previous year-end exchange rate. The pile-up is 2 to 8 percent of nearby firms and 0.06 to 0.30 percent of all firms. It is concentrated among firms whose margin the year before put them on the gaining side of the notch, but firms that reported a loss bunch too, which points to compliance cost. A ceiling enacted in October 2015 for the following year produced within two months a spike just above it, of low-margin firms that then left a regime that was mandatory. Firms moved onto the revenue base by a ceiling increase raise their reported margins, and firms moved off it lower their median margin by 8 to 17 percent, so the fiscal cost of the regime and the revenue gain from narrowing it are both smaller than static estimates imply.

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1 Introduction

Dozens of countries tax small firms on their revenue and larger firms on their profit (Sharma et al., 2025; Kanbur and Keen, 2014; Dharmapala et al., 2011). Best et al. (2015) set out the trade-off: a revenue base is harder to evade and cheaper to administer but taxes costs, a profit base does not but leaves profit to the firm to report. Two questions have stayed open since. What does the base do to what firms report, that is, do firms on profit tax report lower margins than the same firms would on revenue tax? And where does the fiscal cost of a revenue-tax regime sit, with the firms that arrange their revenue to stay under the size limit, or with the intramarginal population that would be inside it anyway? Answering the first needs the same firms observed on both bases without having chosen either, and answering the second needs a measure of behaviour at the limit that does not rest on an assumed shape of the firm-size distribution. The size limit between the regimes is a notch, since crossing it switches the base and, for a firm whose margin is above the ratio of the two rates, raises the tax bill at once, and responses to such notches have been measured at VAT thresholds in the United Kingdom (Liu et al., 2021), Finland (Harju et al., 2019) and Spain (Almunia and López-Rodríguez, 2018), and at turnover-tax and corporate-tax cut-offs in Pakistan (Kleven and Waseem, 2013), Armenia (Asatryan and Peichl, 2016), South Africa (Boonzaaier et al., 2019) and Costa Rica (Bachas and Soto, 2021). In all of these settings the limit is fixed or moves rarely, so the density that would exist without the notch is extrapolated from the density away from it, usually with a polynomial, and Blomquist et al. (2021) show that the result depends on restrictions the data cannot check. Best and Kleven (2018) compare a location before and after a notch was introduced; Marx (2024) uses panel variation.

Romania’s limit moves, and that is what lets us answer both questions. The micro-enterprise regime taxes revenue at 1 to 3 percent for companies whose annual revenue is below a euro ceiling, and the ceiling changed six times: €65,000 until 2015, €100,000 in 2016, €500,000 in 2017, €1,000,000 from 2018 to 2022, €500,000 in 2023 and 2024, €250,000 in 2025 and €100,000 from 2026. Every move reassigned a band of firms between a revenue base and a profit base without any action of their own, which is the experiment the first question needs. And every location that was a notch in one year was an ordinary point of the distribution in others, which supplies a counterfactual without a fitted curve and so a measure of behaviour at the limit for the second.

The paper’s results are these. The pile-up in reported revenue follows the ceiling up and down, and within a fixed euro ceiling it sits within a few thousand lei of the ceiling’s value at the previous year-end exchange rate and moves with it; a spike at 4,779,300 lei that moves to 4,869,400 lei the next year is not a rounding habit. Measured against the same quantile in ceiling-free years, the excess is 2 to 8 percent of the firms within 20 percent of the ceiling, robust to windows, control sets and bootstraps, and centred

on zero at placebo locations; in aggregate it is 0.06 to 0.30 percent of firms. That it is a tax response shows in who does it: bunchers are high-margin firms in professional, administrative, health and information services, the bunching comes mostly from firms whose margin the year before put them on the gaining side of the notch, and a ceiling enacted in October 2015 for the following year produced, within two months, a spike just above it, of low-margin firms that then left a regime that was mandatory at the time. A 2023 rule requiring one employee produced about 20,000 one-employee firms in the year it was enacted and pushed the rest out of the regime. When the 2018 increase moved firms onto the revenue base their reported margins rose relative to firms left on profit tax; when the 2023 cut moved them back, margins fell by 1 to 2 points, 8 to 17 percent of the margin. Because 81 percent of the revenue at stake in the regime comes from firms below €500,000, and because reported profit falls when firms move to the profit base, both the cost of the regime and the gain from narrowing it are smaller than static estimates imply.

Section 2 describes the regime, Section 3 the data, Section 4 the estimator, Section 5 the bunching results, Section 6 the response of reported margins to the base and its fiscal implications, Section 7 the employee condition and two further margins. Section 8 concludes.

2 Institutional background

The micro-enterprise income tax (*impozitul pe veniturile microîntreprinderilor*), Title III of the Fiscal Code, replaces the 16 percent profit tax with a tax on revenue for companies whose revenue in the previous fiscal year was below a ceiling set in euro and that meet side conditions. Table 1 gives both by year of revenue; Appendix A gives the acts.

Three features matter. The first is timing. Under the in-year rule, a micro-enterprise whose cumulated revenue passes the ceiling during year t owes profit tax from that quarter (for 2013 to 2015, for the whole year), the ceiling converted at the central bank (BNR) rate at the close of year $t - 1$. Under the year-end rule, status for year $t + 1$ depends on revenue at 31 December of year t at the rate at the close of year t . When the ceiling is unchanged the two rules put the notch a few thousand lei apart and the in-year rule, at the earlier and lower rate, binds. When a cut has been enacted for the following year a firm faces two notches. That happened in 2022, when Ordinance 16/2022 of July set €500,000 for 2023 status on 2022 revenue and Law 370/2022 of 20 December deferred the revenue test to 2023 revenue; and in 2025, when Ordinance 156/2024 of 31 December 2024 set €250,000 for 2025 and €100,000 for 2026 status on 2025 revenue. A 2016 ceiling of €100,000, adopted on 27 October 2015 and published on 3 November and tested on 2015 revenue, is a further year-end notch that we return to below.

The second feature is the tax base. The ceiling is tested on the taxable base of Article 53, which leaves out items such as subsidies, provision reversals and, from 2021, dividends

Table 1: The micro-enterprise regime by year of revenue

Revenue year	Ceiling (EUR)	Rate on revenue	Conditions
2013–2015	65,000	3%	Mandatory (opt-out for new firms with capital $\geq$ €25,000). Consulting and management revenue at most 20% (from 2014). Exceeding the ceiling in-year re-taxed the whole year.
2016	100,000	1% ($\geq$ 2 employees), 2% (1), 3% (none)	Mandatory. In-year exit from the quarter of crossing.
2017	500,000	1% ($\geq$ 1 employee), 3% (none)	From 1 February. Mandatory, opt-out for capital $\geq$ 45,000 lei. Consulting cap retained.
2018–2022	1,000,000	1% ($\geq$ 1 employee), 3% (none)	Mandatory, opt-out for capital $\geq$ 45,000 lei with two employees (from April 2018). Consulting cap abolished.
2023	500,000	1%	Optional; exit permanent. At least one employee. Consulting cap 20%. A person holding over 25% may be a shareholder in at most three micro-enterprises. Hospitality firms exempt from the ceiling.
2024	500,000	1% up to €60,000; 3% above or in listed activities	At most one micro-enterprise per person holding over 25%. Ceiling tested on linked enterprises combined. Consulting cap reformulated.
2025	250,000	as in 2024	Consulting cap abolished. Status for 2026 tested on 2025 revenue against €100,000.
2026	100,000	1%	Base changed to accounting turnover including linked enterprises.

received. The accounts report total revenue, which is the same or larger; where the excluded items are not zero a firm looks closer to the ceiling than it is, which shifts the bunching region slightly to the right in our data.

The third is the side conditions. The employee condition is met by a full-time contract, part-time contracts adding up to one, or a paid mandate contract; linked enterprises are defined by holdings above 25 percent; the activities taxed at 3 percent regardless of revenue in 2024 and 2025 are software, hospitality, legal and medical services. The consulting cap, in force in 2014–2017 and, reformulated, in 2023–2024, is a margin the accounts do not show. The VAT exemption for small firms is a separate notch in lei, 220,000 lei until March 2018, 300,000 from 1 April 2018 and 395,000 from 1 September 2025, which Appendix C treats.

3 Data

The Ministry of Finance publishes, for each fiscal year, the main figures from the annual accounts of every company, identified by fiscal code (CUI) and CAEN activity code: balance-sheet totals, net turnover, total revenue, total expenses, gross and net profit or loss, and average headcount. The layout is stable from 2016; the 2014 and 2015 files use an

Table 2: Statement panel, 2014–2025

Year	Firm-years	Revenue > 0	Median revenue (k lei)	EUR/RON, 31 Dec
2014	676,868	495,914	124	4.4821
2015	681,821	505,212	141	4.5245
2016	700,249	526,282	146	4.5411
2017	732,051	553,283	151	4.6597
2018	757,929	579,309	162	4.6639
2019	792,376	616,900	173	4.7793
2020	815,779	636,361	162	4.8694
2021	781,470	624,795	188	4.9481
2022	862,564	693,309	207	4.9474
2023	890,836	716,718	223	4.9746
2024	968,594	769,620	226	4.9741
2025	918,232	738,762	239	5.0985

Source: Ministry of Finance annual financial statements (micro-entity and full forms) via data.gov.ro; BNR year-end reference rate. Median revenue is over firms with positive revenue. 32 duplicated firm-years (30 in 2016) removed; about 600 firm-years per year with negative revenue are excluded from all windows.

older set of columns, which we harmonise. The 2021, 2023 and 2025 files were published a few weeks after the filing deadline and the others are later re-issues, so late filers are covered unevenly. Table 2 summarises the panel: 9.58 million firm-years after removing 32 duplicates, 1.56 million firms. Total revenue is the variable for the micro-enterprise notch.

The accounts do not say which regime a firm is in. We recover it in two ways. Gross profit minus net profit is the income tax accrued; divided by revenue it is roughly 1, 2 or 3 percent for micro-enterprises, and divided by pre-tax profit it is roughly 16 percent for profit-tax firms. The ANAF taxpayer registry, a snapshot dated 31 August 2023, has one column per tax a firm is registered for. Table 3 cross-tabulates the two for 2023: of firms registered for the micro-enterprise tax, 84 percent show a micro pattern and 1 percent a profit-tax pattern; of firms registered for profit tax, 22 percent show the profit-tax pattern and half paid no tax. Where both are decisive, 57 percent of firms, they agree in 92.5 percent of cases. The inference picks out micro-enterprises well and profit-tax firms poorly, so we use the registry flag where membership matters and the accounts-based measure as a check.

Headcount is blank for a quarter of small firms; we treat blanks as zero, and among micro-enterprises three quarters of blank-headcount firms paid the no-employee rate. The field is a whole number and behaves as a flag for having had an employee during the year rather than a time-weighted average, so late hires are counted. The trade register gives each firm’s registration date, address and administrators as of the snapshot of 2 September 2026; persons are keyed on name, birth date and birth place and hashed.

Table 3: Tax regime inferred from the 2023 statements versus the ANAF tax-vector flag (snapshot of 31 August 2023)

Regime inferred from statements	ANAF: micro	ANAF: profit	ANAF: other/none	Not in ANAF
Micro (tax $\approx$ 1–3% of revenue)	277,265	19,677	312	7,102
Micro (tax $>$ 0, pre-tax loss)	103,544	14,248	87	1,966
Profit tax ($\approx$ 16% of profit)	4,421	92,503	64	2,045
No tax paid	34,486	220,021	273	10,942
Ambiguous	16,047	30,596	60	437
Other	19,272	34,966	158	344

ANAF flags: IMP120 = registered for micro-enterprise income tax, IMP100 = registered for profit tax. Where both measures are decisive, they agree for 92.5% of firms.

Administrators, not shareholders, are our proxy for ownership; 85 percent of the limited-liability companies in the sample have exactly one. Appendix B gives details. Exchange rates are BNR reference rates at the last quotation of each year.

4 Empirical strategy

For each year t the notch location is $L_t = T_t e_{t-1}$, where T_t is the euro ceiling for revenue in year t and e_{t-1} the BNR rate at the close of year $t - 1$. Let F_t be the distribution of revenue in year t and $q_t = F_t(L_t)$ the quantile of the notch; for every other year c let $L_c^* = F_c^{-1}(q_t)$ be the revenue at the same quantile. Year c is an admissible control if no known notch of year c , the micro ceiling, an enacted or announced future ceiling, the €60,000 rate boundary, or the VAT threshold, lies within 25 percent of L_c^* . Bins are 1 percent of the location wide in a window of ± 20 percent; shares $s_{t,k}$ are bin counts divided by the count in the window, and the counterfactual share is the mean of $s_{c,k}$ over admissible controls. Excess mass is

$$b_t = \frac{\sum_{k \in B} (s_{t,k} - \bar{s}_{c,k})}{\frac{1}{|B|} \sum_{k \in B} \bar{s}_{c,k}}, \quad (1)$$

with B the five bins just below the notch; m_t is the same sum over the five bins just above. With 1 percent bins b is roughly the width of the bunching region in percent of the ceiling. The identifying assumption is that the shape of the revenue density around a given quantile is the same across years apart from the notch. Quantile matching makes the counterfactual immune to inflation and to growth in firm size, not to changes in shape, so we test it: the estimator is run at 74 placebo locations, at 0.45 to 2.5 times each year’s ceiling and at least 25 percent from any notch, where the true excess is zero. The notch list includes the year-end ceilings enacted for the following year, one of which, the €100,000 ceiling for 2016 tested on 2015 revenue, we found because an earlier placebo grid built without it had its largest value at that location. Because shares are normalised within the window, the excess and missing mass offset by construction; we

Table 4: Location of the revenue spike in lei, 2018–2022

Year	€1M, prev. rate (lei)	Modal 5,000-lei bin	Firms	Bin centre – threshold	Current – prev. rate
2018	4,659,700	4,655,000-4,660,000	59	-2,200	+4,200
2019	4,663,900	4,660,000-4,665,000	73	-1,400	+115,400
2020	4,779,300	4,775,000-4,780,000	95	-1,800	+90,100
2021	4,869,400	4,865,000-4,870,000	105	-1,900	+78,700
2022	4,948,100	4,940,000-4,945,000	101	-5,600	-700

Modal 5,000-lei bin within 60,000 lei of either candidate threshold. The last column gives the lei value of €1M at the current year-end rate minus its value at the previous year-end rate (the dotted line in Figure 1). At 1,000-lei resolution the modal bins hold only 16 to 31 firms and are noisier; they lie 2,000 to 5,000 lei below the previous-rate threshold in 2020–2022 and further below in 2018–2019.

therefore also compute a count-based version in which each bin is a share of all firms in the year, N_t , so that the integration constraint becomes a test of where bunchers come from. Uncertainty is reported three ways: a multinomial bootstrap of bin counts, a block bootstrap that resamples control years, and the placebo distribution. Appendix Table 22 lists every control year with its matched revenue and distance to the nearest notch; the closest cases are 26 to 34 percent, for the 2014–2016 notches and the two year-end notches, and tightening the rule to 35 percent changes the estimates from 2016 on by at most 0.01.

Three things the estimator does not deliver are stated once here. It does not locate the missing mass, which is spread over the window above the notch rather than concentrated next to it, so the standard conversion of excess mass to an elasticity has no anchor. The tax change on crossing depends on each firm’s margin, because the base switches, so a representative-firm elasticity has no clear meaning. And reported margins themselves respond to the base, as Section 6 shows. Appendix Table 26 reports the Kleven–Waseem reduced-form elasticity under two proxies for the bunching width, which differ by three orders of magnitude; we report excess mass and excess shares instead.

5 Bunching at the ceiling

5.1 The spike follows the ceiling and the exchange rate

Figure 2 shows a spike in the bin just below the ceiling in every year: at €65,000 in 2014 and 2015, €100,000 in 2016, €500,000 in 2017, €1,000,000 from 2018 to 2022, €500,000 in 2023 and 2024 and €250,000 in 2025. Figure 1 looks inside the five years at €1,000,000, when the ceiling’s lei value rose from 4.66 to 4.95 million as the leu depreciated. The spike moves with it each year; Table 4 puts the modal 5,000-lei bin 1,400 to 5,600 lei below the ceiling at the previous year-end rate, the rate the in-year rule prescribes, while the ceiling at the current year-end rate is 79,000 to 115,000 lei further right in 2019 to 2021. Nobody rounds revenue to 4,779,300 lei, and a spike at that value that moves to 4,869,400 lei the next year is not an accounting convention.

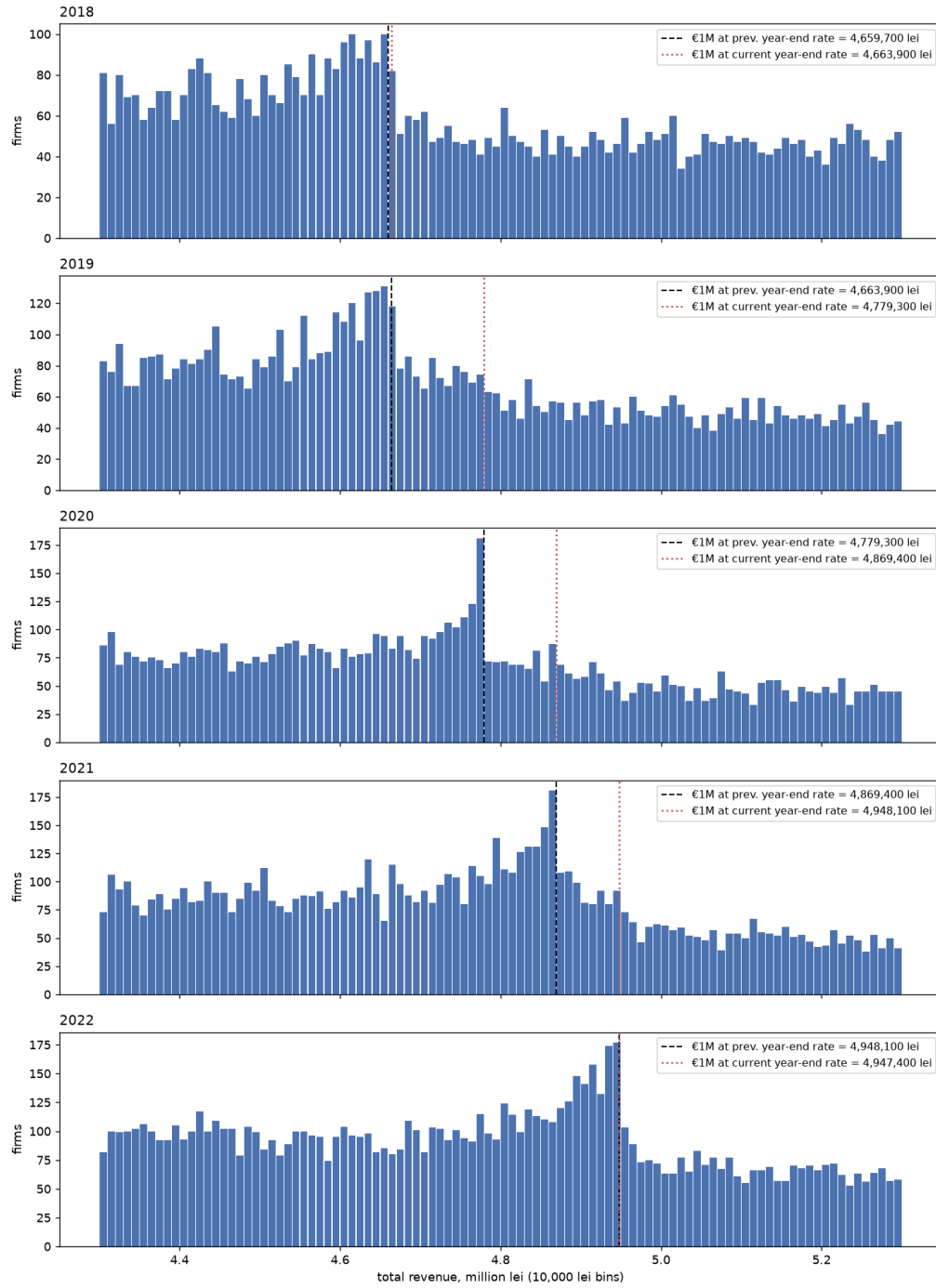


Figure 1: Total revenue in lei around €1,000,000, 2018–2022, 10,000-lei bins. Dashed: ceiling at the previous year-end rate; dotted: at the current year-end rate.

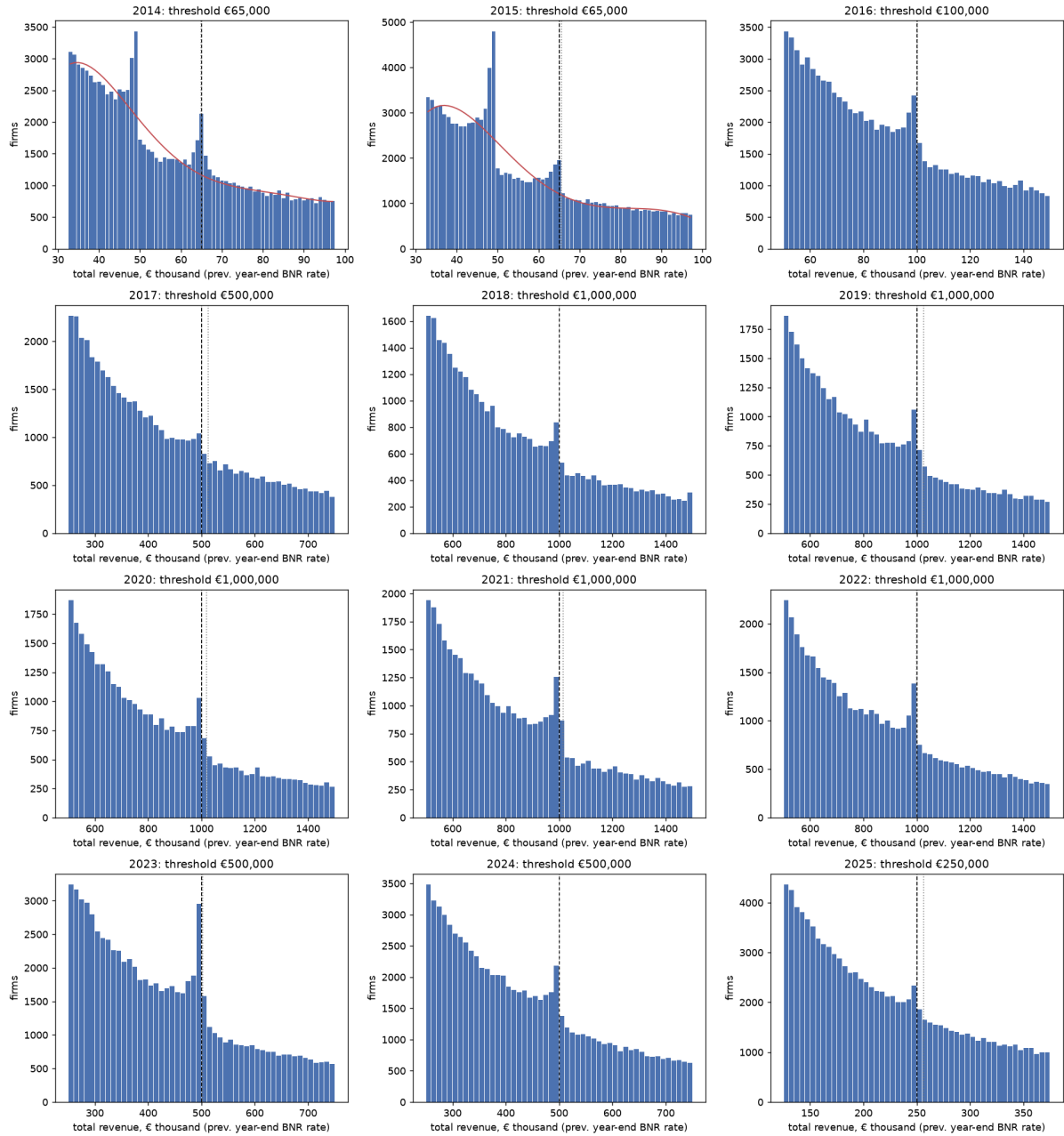


Figure 2: Total revenue around the micro-enterprise ceiling, 2014–2025, converted at the BNR rate at the close of the previous year. Dashed: ceiling; dotted: ceiling at the current year-end rate. In 2014 and 2015 the taller spike at about €49,000 is the VAT threshold of 220,000 lei; the red line there is a polynomial counterfactual, shown because it passes through that spike.

5.2 Difference-in-bunching estimates

Table 5 reports the estimates and Figure 3 the densities for four notches; all fifteen are in Appendix Figure 6. In every year the treated density exceeds the same-quantile control density in the bins just below the notch, and the control density is smooth through it. Excess mass in the 5 percent below the notch ranges from 1.9 percent of firms in the window, at the €500,000 ceiling of 2017 and the €250,000 ceiling of 2025, to 7.6 percent at the €500,000 ceiling of 2023. As a share of all filers it is 0.06 to 0.30 percent. Table 6

shows that a block bootstrap over control years gives standard errors of 0.08 to 0.16, that dropping any one control year moves b by at most 0.1, and that narrowing the window to ± 10 percent lowers b by 15 to 36 percent and the excess count by 12 to 31 percent, while widening it to ± 30 percent raises both by about a tenth or less, except in 2014 where the VAT spike enters the wider window. That dependence is mechanical: shares are normalised within the window, so a narrower window deflates every treated share by the bunchers' own weight in it, and the count-based excess of Table 7, which does not use the window, is identical to within three firms at all three widths; Appendix Table 23 adds that control years before and after the treated year agree, that correcting the control quantile for the bunchers changes b by at most 0.1, and that continuing firms give b lower by 0.05 to 0.2.

The placebo distribution is the relevant benchmark. Across the 74 locations b has mean 0.01 and standard deviation 0.13; the 95th and 99th percentiles of $|b|$ are 0.27 and 0.34 and the maximum is 0.39. Every in-year estimate exceeds that maximum, most by a factor of two or more; the 2017 and 2025 in-year estimates, 0.76 and 0.73, are the weakest, and in the count-based version, where the placebo maximum is 0.61, they are 0.94 and 1.00, only one and a half times it. Two things dilute the 2025 notch: the enacted €100,000 ceiling for 2026 competed for attention, and at the 3 percent rate then in force the break-even margin was above the median buncher's. A within-year placebo points the same way as the grid: hospitality firms, exempt from the ceiling in 2023, show an excess of under 1 percent of their window that year against 7.6 percent for all firms (Table 10).

Table 7 gives the count-based version. At every large notch the treated year has more firms than the counterfactual across the whole 20 percent below the notch, not only in the five bins next to it, and a deficit in the 20 percent above of at most a seventh of the excess; the net difference over the window is 10 to 17 percent of the window for the 2019 to 2024 notches, against a placebo 95th percentile of 5 percent. Bunchers are therefore not drawn from the region just above the ceiling, and the width of the bunching region, and any elasticity derived from it, is not identified in these data. Where the extra firms come from is less settled. Three candidates: firms below the ceiling grow less, so the density below it accumulates over the years the ceiling is in place, the build-up that Kleven and Waseem (2013) document for Pakistan and that shows here as a rise from 3.5 percent of the window in 2018 to 5.5 percent in 2021; new and split firms enter the window, 11 percent of the 2023 excess being firms in their first year; or the shape assumption holds in shares but not in counts. Two checks are in Appendix Tables 27 and 28. Extending the window to ± 50 percent finds, for 2019 to 2024, a surplus of 0 to 5 percent of the counterfactual in the first 10 percent above the notch, deficits of 1 to 8 percent in the 10 to 30 percent above and nothing systematic further out, but also a surplus of 2 to 8 percent from 20 to 50 percent below, further than a bunching response reaches, and for the 2014 to 2016 notches a deficit of 11 to 17 percent there; the count-based comparison picks

Table 5: Difference-in-bunching estimates

Notch	Location (M lei)	Firms in window	b (SE)	m	Excess firms	% of window	Range over 3–8% regions
2014: in-year, €65,000	0.292	33,859	1.15 (0.08)	0.77	986	2.9	931–986
2015: in-year, €65,000	0.291	34,896	1.71 (0.09)	-0.39	1,526	4.4	1,232–1,674
2016: in-year, €100,000	0.452	32,891	1.64 (0.09)	-0.26	1,369	4.2	1,036–1,540
2017: in-year, €500,000	2.271	17,172	0.76 (0.11)	-0.26	332	1.9	262–370
2018: in-year, €1,000,000	4.660	11,424	1.41 (0.16)	-0.59	406	3.5	341–467
2019: in-year, €1,000,000	4.664	13,105	1.67 (0.15)	0.05	558	4.3	494–605
2020: in-year, €1,000,000	4.779	12,693	1.75 (0.14)	-0.01	576	4.5	473–655
2021: in-year, €1,000,000	4.869	14,394	2.15 (0.13)	-0.08	793	5.5	670–916
2022: in-year, €1,000,000	4.948	16,448	1.92 (0.13)	-0.45	808	4.9	711–829
2023: in-year, €500,000	2.474	28,379	2.99 (0.11)	-0.14	2,151	7.6	1,850–2,388
2024: in-year, €500,000	2.487	28,619	1.62 (0.10)	-0.28	1,179	4.1	945–1,339
2025: in-year, €250,000	1.244	37,048	0.73 (0.08)	-0.06	690	1.9	588–697
2015: year-end, €100,000	0.452	29,480	-0.03 (0.07)	0.71	-24	-0.1	-89 to -19
2022: announced, €500,000	2.474	24,241	0.80 (0.09)	-0.13	494	2.0	359–531
2025: year-end, €100,000	0.497	55,232	1.47 (0.07)	0.47	2,059	3.7	1,616–2,278

Location = EUR threshold $\times$ BNR rate at the close of the previous year (in-year notches) or of the same year (the 2015 year-end notch; the 2025 year-end notch at the end-2024 rate, the last rate firms could observe). Window $\pm 20\%$, bins 1% of the location. Control years are listed in Appendix Table 22. b = excess mass in the five bins below the notch in units of counterfactual firms per bin; m = the same over the five bins above, negative meaning a deficit (a hole) and positive an excess; SE from 300 multinomial bootstrap draws. Excess firms = excess in the 5% region below the notch, with its range over 3% and 8% regions.

Table 6: Robustness of the difference-in-bunching estimates

Notch	b	Block-bootstrap SE	b , window $\pm 10\%$	$\pm 30\%$	Excess firms, window ± 10 / 20 / 30%	Leave-one-out range	B/N_t (%)
2014: in-year, €65,000	1.15	0.08	0.74	0.78	680 / 986 / 708	1.13–1.18	0.20
2015: in-year, €65,000	1.71	0.08	1.36	1.55	1,278 / 1,526 / 1,417	1.69–1.72	0.30
2016: in-year, €100,000	1.64	0.09	1.28	1.79	1,134 / 1,369 / 1,462	1.62–1.66	0.26
2017: in-year, €500,000	0.76	0.11	0.62	0.79	278 / 332 / 343	0.73–0.79	0.06
2018: in-year, €1,000,000	1.41	0.14	1.20	1.46	357 / 406 / 418	1.38–1.43	0.07
2019: in-year, €1,000,000	1.67	0.14	1.27	1.87	453 / 558 / 607	1.63–1.71	0.09
2020: in-year, €1,000,000	1.75	0.16	1.35	1.93	472 / 576 / 617	1.71–1.83	0.09
2021: in-year, €1,000,000	2.15	0.15	1.59	2.39	637 / 793 / 853	2.10–2.20	0.13
2022: in-year, €1,000,000	1.92	0.14	1.58	2.15	700 / 808 / 876	1.87–1.97	0.12
2023: in-year, €500,000	2.99	0.10	2.19	3.35	1,749 / 2,151 / 2,304	2.97–3.02	0.30
2024: in-year, €500,000	1.62	0.09	1.26	1.78	967 / 1,179 / 1,262	1.58–1.64	0.15
2025: in-year, €250,000	0.73	0.08	0.57	0.78	557 / 690 / 728	0.71–0.75	0.09
2015: year-end, €100,000	-0.03	0.08	-0.14	0.04	-106 / -24 / 28	-0.05 to -0.01	-0.01
2022: announced, €500,000	0.80	0.09	0.60	0.89	384 / 494 / 538	0.78–0.82	0.07
2025: year-end, €100,000	1.47	0.07	1.02	1.11	1,540 / 2,059 / 1,641	1.44–1.50	0.28

Baseline: window $\pm 20\%$, all admissible control years. Block-bootstrap SE: control years resampled with replacement and treated bin counts resampled, 500 draws. Excess firms: treated minus counterfactual firms in the 5% region below the notch when shares are normalised within each window. Leave-one-out: range of b dropping one control year at a time. B/N_t : excess firms in the 5% region as a share of all filers with positive revenue in year t . Further checks in Appendix Table 23.

up differences in the shape of the distribution across years as well as the notch, which is what the window normalisation removes. The growth check is closer to the mechanism. In every notch year in which the ceiling was unchanged the following year, firms 5 to 20 percent below the notch grew less from t to $t+1$, relative to firms 20 to 35 percent below, than the same bands did in control years, by 0.3 to 2.6 points, and in four of the six years the share crossing the next year’s notch was 3 to 9 points lower. Slower growth below the ceiling is there, but it is small, and the frictions reading of Kleven and Waseem (2013), a region above the notch where most firms do not move, fits the flat density above it as well. We read the surplus as partly accumulation and partly shape drift, and the conclusions do not rest on it.

Two further features of Table 5. The 2023 response is the largest and falls to 4.1

Table 7: Count-based difference-in-bunching: firms relative to the same-quantile counterfactual scaled to the year's filers

Notch	b , count-based	Excess, 5 bins below	5 bins above	5-20% above	5-20% below	Net, whole window	Net, % of window
2014: in-year, €65,000	1.23	1,043	+680	+41	-620	+464	1.4
2015: in-year, €65,000	1.75	1,554	-301	-1,446	+120	+228	0.7
2016: in-year, €100,000	1.95	1,557	-29	-766	+692	+1,483	4.5
2017: in-year, €500,000	0.94	399	-44	-119	+245	+525	3.1
2018: in-year, €1,000,000	1.74	477	-93	-212	+300	+565	4.9
2019: in-year, €1,000,000	2.46	736	+176	-66	+721	+1,390	10.6
2020: in-year, €1,000,000	2.50	739	+143	-104	+627	+1,262	9.9
2021: in-year, €1,000,000	3.35	1,057	+213	-2	+1,007	+2,062	14.3
2022: in-year, €1,000,000	2.81	1,049	+48	-29	+861	+1,882	11.4
2023: in-year, €500,000	4.59	2,749	+460	-89	+2,064	+4,723	16.6
2024: in-year, €500,000	2.38	1,551	+157	-42	+1,425	+2,934	10.3
2025: in-year, €250,000	1.00	907	+152	+39	+762	+1,708	4.6
2015: year-end, €100,000	-0.20	-155	+377	+52	-919	-1,022	-3.5
2022: announced, €500,000	0.97	580	+5	-303	+399	+676	2.8
2025: year-end, €100,000	2.29	2,850	+1,363	+649	+2,752	+6,250	11.3

Each bin is a share of all firms with positive revenue in the year rather than of the window, so the counterfactual is not forced to integrate to the treated total. Columns give treated minus counterfactual in firms. At 74 placebo locations the net window difference has mean -0.5% of the window, standard deviation 3.1 and 95th percentile of the absolute value 5.4; the count-based b has standard deviation 0.19 and maximum 0.61.

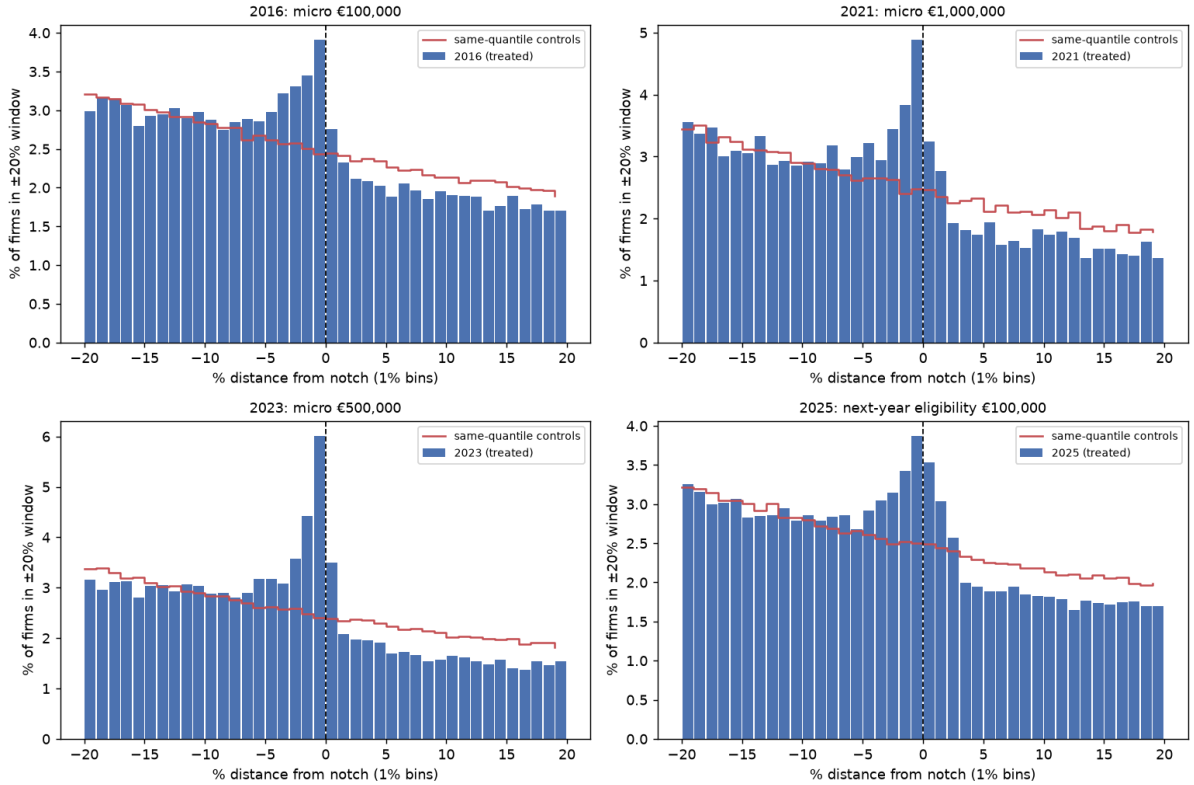


Figure 3: Treated-year density (bars) and same-quantile control density (line) around four notches, bins of 1 percent of the notch location.

percent in 2024 at the same ceiling; the ceiling halved and exit became permanent that year. The share of 2023 bunchers that had been above €500,000 in 2022, 29.6 percent, is no higher relative to a comparison group than in years when that point was not a ceiling (Appendix Table 24), so it is mean reversion, not displaced firms requalifying. And the three extra rows show year-end notches operating, each with a different timing. In 2015 the €100,000 ceiling for 2016 status, enacted in October 2015 with two months of the year left, produced no excess below its lei value at the end-2015 rate, 452,450 lei, but an excess of 0.71 (standard error 0.08) in the five bins above it, about 500 firms or 1.7 percent of the window, most of them in the first two bins; the 2014 distribution is flat there. The regime was mandatory, so a firm below the ceiling that expected to lose from a 1 to 3 percent tax on revenue in 2016 could avoid it only by having 2015 revenue above the ceiling. Appendix Table 30 follows the two sides of the ceiling into 2016. Firms 0 to 3 percent above it were on profit tax the next year, 21 percent with a profit-tax pattern and 32 percent paying no tax, against 9 and 13 percent for firms 0 to 5 percent below, of which 57 percent show the micro pattern;¹ the median tax paid in 2016 was 0.5 percent of revenue above the ceiling and 1.0 percent below. The firms in the spike have low margins, a median of 3.8 percent with 60 percent below the 6.25 percent break-even (Appendix Table 29), so they gained from being above. What the panel does not show is a reversal: their 2016 revenue grew by a median 4.2 percent, no less than the 2.5 percent of firms 3 to 10 percent above, and the share that fell back below the ceiling's value, 42 percent, is within a few points of that group and of the 2014 placebo cohort. One detail of the location supports the reading. At the rates visible while the year was still open, monthly averages of 4.42 lei in October, 4.44 in November and 4.50 in December 2015 (Banca Națională a României, 2015), €100,000 was worth 442,000 to 450,000 lei; the spike does not start there but at 452,000, just above the end-year value of 452,450 that fixed the test. Firms cleared the ceiling at its year-end value, whether by leaving a margin for the rate or by placing revenue once the rate was known, and neither round-number nor fixed-lei targeting produces a spike that begins at 452,450 lei. The spike is firms that placed themselves on the profit-tax side of a ceiling they expected to lose from and stayed there; whether the placement was a timing of real revenue or a reporting choice the data cannot tell. In 2022 a spike of 2.0 percent of the window appears at €500,000, the ceiling firms expected for 2023 status until the December deferral, with none at that location in 2021. In 2025 a spike of 3.7 percent appears at €100,000, the ceiling enacted for 2026. That spike runs from 494,000 to 500,000 lei at 1,000-lei resolution, straddling the €100,000

¹A fifth of the firms above the ceiling also show a micro pattern in 2016, when the law put them on profit tax. We read this as inference noise rather than as Article 53 exclusions moving their taxable base below the ceiling: in the 2014 cohort at the same quantile, where every firm was above the €65,000 ceiling and on profit tax, 14 to 15 percent show the same pattern, because a profit-tax firm with a margin of 6 to 19 percent pays 16 percent of profit, which is 1 to 3 percent of revenue, the signature the inference reads as the revenue tax. The exclusions may account for the remaining few points.

value at the end-2024 rate, 497,410 lei, which is why m is positive for that row; the data cannot separate targeting of the last observable rate from targeting of 500,000 lei as a round number. For comparison, a polynomial counterfactual of the kind used by Chetty et al. (2011), at bin widths of 1.5 to 2 percent of the ceiling and with the VAT spike inside the 2014 and 2015 windows, gives 0.4 to 2.0 at baseline and, across 27 combinations of bin width, degree and excluded window, anything from negative values to more than twice the baseline; the two sets of estimates are not on the same scale and the comparison is only about stability.

5.3 Who bunches

The notch raises taxes only for firms whose margin exceeds the ratio of the two rates, 6.25 percent at a 1 percent revenue tax and 18.75 percent at 3 percent. Table 8 compares firms in the 5 percent below the ceiling with firms just above it, in the notch year and in the year before. The year before matters because Section 6 shows that the base changes what firms report: firms just above the ceiling in year t crossed it during the year and closed their accounts on profit tax, so part of any gap in year- t margins is the base effect rather than selection. In $t - 1$ the two groups were on more similar bases, in 2017 and 2018 nearly all on profit tax because the previous ceiling was lower, and the gap survives: bunchers' median margins in $t - 1$ are 7 to 20 percent against 3 to 12 percent for firms just above, and the share above the break-even margin in $t - 1$ is 40 to 76 percent against 23 to 66. Table 10 shows where the bunchers are: in 2023, excess mass is 11 to 15 percent of the window in professional, administrative, health and real-estate services and in construction, and 10 percent in information services, against 5 percent in trade, 3.5 in transport and 1 in agriculture, and bunchers in the service sectors report margins of 29 to 43 percent.

Table 9 makes the split the unit of estimation. Firms are divided by their $t - 1$ margin into those above the break-even margin for the rate at the ceiling and those at or below it, and the estimator is run within each group, each matched at its own quantile in the same control years. Bunching is concentrated where the tax wedge is: among firms that gain, b is 1.3 to 4.4 and the excess 3 to 11 percent of the window; among firms with a positive margin at or below break-even, b is 0 to 1.3 and the excess 0 to 3 percent. This resolves the puzzle of the 3 percent years, in which the median buncher's margin, 13 to 17 percent, is below the 18.75 percent at which a 3 percent revenue tax saves tax, yet 2014 and 2015 show excess mass of 1.2 and 1.7. In 2015 the gaining group, a quarter of firms, has b of 4.4 and an excess of 11 percent of its window, the low-positive group 1.3 and 3 percent; in 2024 and 2025 the figures are 3.4 and 1.5 against 0.6 and 0.2. The median hides a mix, and most of the bunching comes from the minority that gains. The third group is the informative one. Firms that reported a loss in $t - 1$, for whom crossing the ceiling removes the tax altogether, bunch more than firms with a small positive margin:

Table 8: Pre-tax profit margins by position relative to the ceiling, in the notch year and the year before

Year	Median margin t : bunchers	just above	Median margin $t - 1$: bunchers	just above	Bunchers above break-even, t (%)	Above break-even, $t - 1$: bunchers (%)	just above (%)	Rate paid (%)
2014	0.160	0.027	—	—	47	—	—	3.0
2015	0.169	0.058	0.108	0.034	48	41	23	3.0
2016	0.134	0.053	0.089	0.044	63	56	43	1.0
2017	0.089	0.062	0.068	0.051	58	52	45	1.0
2018	0.106	0.057	0.077	0.053	62	55	45	1.0
2019	0.117	0.070	0.101	0.067	65	61	52	1.0
2020	0.132	0.078	0.109	0.067	69	64	51	0.9
2021	0.160	0.097	0.134	0.082	74	69	57	0.9
2022	0.166	0.101	0.156	0.105	75	73	65	0.9
2023	0.200	0.098	0.201	0.121	76	76	66	0.9
2024	0.151	0.081	0.154	0.099	45	45	32	2.9
2025	0.135	0.093	0.121	0.098	43	40	35	2.7

Bunchers: 0–5% below the ceiling in year t ; just above: 0–10% above. Break-even margin: 6.25% where the rate at the ceiling was 1% and 18.75% where it was 3% (2014, 2015, 2024, 2025). Margins in $t - 1$ are for the same firms one year earlier, when firms just above had not yet crossed in-year. Last column: median of accrued tax over revenue among bunchers inferred to be on the revenue tax; in 2016 the rate depended on headcount and the median buncher paid 1%.

Table 9: Difference-in-bunching within groups defined by the margin reported in $t - 1$

Notch	Break-even (%)	Gain: b (SE)	excess, % of window	Low positive: b (SE)	excess, % of window	Loss: b (SE)	excess, % of window
2015: in-year, €65,000	18.75	4.44 (0.20)	11.3	1.27 (0.14)	3.0	1.24 (0.18)	2.8
2016: in-year, €100,000	6.25	2.56 (0.14)	6.4	0.32 (0.15)	0.8	0.69 (0.16)	1.8
2017: in-year, €500,000	6.25	1.34 (0.19)	3.4	-0.00 (0.17)	-0.0	0.60 (0.28)	1.5
2018: in-year, €1,000,000	6.25	2.02 (0.23)	5.2	0.49 (0.22)	1.2	0.73 (0.43)	1.8
2019: in-year, €1,000,000	6.25	2.31 (0.23)	6.0	0.51 (0.21)	1.3	1.56 (0.42)	3.9
2020: in-year, €1,000,000	6.25	2.66 (0.23)	6.9	0.41 (0.20)	1.1	0.92 (0.45)	2.3
2021: in-year, €1,000,000	6.25	3.17 (0.21)	8.1	0.33 (0.24)	0.8	1.03 (0.38)	2.6
2022: in-year, €1,000,000	6.25	2.49 (0.19)	6.4	0.18 (0.24)	0.5	1.55 (0.45)	3.8
2023: in-year, €500,000	6.25	3.62 (0.14)	9.2	0.79 (0.19)	2.0	2.02 (0.34)	5.0
2024: in-year, €500,000	18.75	3.38 (0.22)	8.6	0.59 (0.13)	1.5	0.58 (0.23)	1.5
2025: in-year, €250,000	18.75	1.53 (0.14)	3.8	0.19 (0.11)	0.5	0.33 (0.18)	0.8

Gain: firms whose margin in $t - 1$ exceeded the break-even margin for the rate at the ceiling (25–63% of firms with a $t - 1$ statement, by year); low positive: margin in $t - 1$ above zero and at or below break-even (15–37%); loss: margin in $t - 1$ at or below zero (22–38%). Firms without a $t - 1$ statement are excluded. Control-year groups are formed by the same $t - 1$ margin rule at the treated year’s break-even, so the shape assumption is made within group; each group is matched at its own quantile in the same control years as the pooled estimate. SE from 200 multinomial bootstrap draws.

b of 0.3 to 2.0 and an excess of 0.8 to 5 percent of the window, two or more standard errors from zero in nine of eleven years. Some of them will have expected a profit their previous year did not show, but for the rest no tax wedge holds them below the ceiling, and this is the cleanest evidence in the paper for the mechanism Harju et al. (2019) find at the Finnish VAT threshold, compliance cost: leaving the regime meant, in those years, moving from a one-line tax base to quarterly profit computations, the deductibility rules of the profit tax and the audit exposure that comes with them. In 2016 the rate depended on headcount and the median buncher paid 1 percent, so 2016 is not a 3 percent year.

Table 10: Excess mass by sector, in percent of firms within 20% of the notch

CAEN section	2021 (€1M)	2023 (€500k)	2025 (€250k)	Firms in 2023 window	Median margin of 2023 bunchers
Professional services	–	14.8	4.6	2,068	0.39
Administrative services	–	13.3	3.4	1,089	0.29
Construction	7.4	12.6	3.1	4,078	0.25
Health	–	12.0	2.6	896	0.32
Real estate	–	11.2	3.5	908	0.43
Information and communication	–	10.1	5.1	1,124	0.34
Manufacturing	2.9	6.5	1.9	3,008	0.16
Trade	5.3	4.8	0.3	8,426	0.12
Transport	6.1	3.5	-0.2	2,428	0.10
Agriculture	0.5	1.0	1.0	1,653	0.08
Hospitality	–	0.9	0.4	1,571	0.14
Arts, other services	–	–	1.7	–	–

Difference-in-bunching within each CAEN section, matched at the section’s own quantile, same control years as the pooled estimate; sections with fewer than 800 firms in the window omitted. Hospitality faced no revenue ceiling in 2023.

6 The base changes what firms report

A micro-enterprise’s tax does not depend on its reported profit; a profit-tax firm’s does. Firms on profit tax therefore have a reason to report lower margins, by booking costs aggressively, or by paying the owner a salary, which is deductible under profit tax and taxed on the owner, rather than a dividend, which is not deductible and was taxed at 5 to 8 percent. Both responses lower reported profit; only the first is misreporting. The accounts do not separate personnel costs, so we cannot tell them apart and say so; what we can do is measure the total.

6.1 Design

In 2018 the ceiling rose from €500,000 to €1,000,000, so firms with 2017 revenue between those values were moved from profit tax to revenue tax without doing anything, while firms above €1,000,000 stayed on profit tax. In 2023 the cut reversed the move. Wide bands, (€500,000, €1,000,000] against (€1,000,000, €2,000,000], compare firms that differ twofold in size and produce mean-reversion drift of up to 1.5 points in a placebo (Appendix Table 25). We therefore use bands 20 percent either side of the cut-off, assigned on two-year average revenue, in balanced panels, and placebos built at the same time with the same bands at €1,500,000 and €2,000,000, where no regime changed. Table 11 gives the first stage. In 2018 the treated band’s micro share, inferred from the accounts, jumps from 15 to 55 percent; the control band just above the new ceiling goes from 15 to 29 percent, because some of those firms bunched under it, so the control is partly treated, which is why the entry effect fades over the horizon in the baseline. A control band of 1.2 to 1.5 times the cut-off stays at 15 to 25 percent. The exit design is fuzzier and its first stage is not well measured. The inference classifies 42 percent of the control band as micro in 2022, above the ceiling, because firms that cross the ceiling during the year pay both taxes and show a blended ratio. The ANAF registry, which records the regime

Table 11: First stage of the margin design

Year	Micro share, treated (%)	Micro share, control (%)	First stage (pp)	DiD, median margin (pp)	DiD, mean margin (pp)	Per switch, means (pp)	n T / C
<i>2018 entry, control band $(1.0, 1.2] \times \text{€}1\text{M}$</i>							
2015	14.2	12.8	+0.6	+0.09	+0.72	– 5,353 / 3,760	
2016	13.8	13.6	-0.6	+0.16	+0.35	– 5,353 / 3,760	
2017	15.3	14.5	+0.0	+0.00	–	– 5,353 / 3,760	
2018	55.5	29.2	+25.5	+0.71	+0.94	+3.67 5,353 / 3,760	
2019	45.3	28.8	+15.8	+0.51	+0.69	+4.34 5,353 / 3,760	
2020	44.1	32.8	+10.5	+0.48	+0.60	+5.71 5,353 / 3,760	
<i>2018 entry, control band $(1.2, 1.5] \times \text{€}1\text{M}$</i>							
2015	14.2	13.5	-0.1	-0.09	+0.42	– 5,353 / 3,807	
2016	13.8	12.8	+0.2	-0.26	+0.16	– 5,353 / 3,807	
2017	15.3	14.5	+0.0	+0.00	–	– 5,353 / 3,807	
2018	55.5	17.8	+36.9	+0.60	+0.85	+2.32 5,353 / 3,807	
2019	45.3	21.1	+23.4	+0.34	+0.51	+2.17 5,353 / 3,807	
2020	44.1	25.4	+18.0	+0.52	+0.46	+2.57 5,353 / 3,807	
<i>2023 exit, control band $(1.0, 1.2] \times \text{€}1\text{M}$</i>							
2020	69.1	58.9	-14.4	-0.06	-0.16	– 7,556 / 4,766	
2021	70.2	57.3	-11.8	+0.07	-0.07	– 7,556 / 4,766	
2022	66.3	41.7	+0.0	+0.00	+0.00	– 7,556 / 4,766	
2023	40.1	24.9	-9.4	-0.95	-1.36	– 7,556 / 4,766	
2024	19.3	18.5	-23.8	-1.62	-2.51	– 7,556 / 4,766	
2025	17.5	16.8	-24.0	-1.98	-3.04	– 7,556 / 4,766	
2023, ANAF flag	39.7	12.4	–	–	–	– 7,556 / 4,766	
2024, ANAF flag	6.0	2.2	–	–	–	– 7,556 / 4,766	
<i>2023 exit, control band $(1.2, 1.5] \times \text{€}1\text{M}$</i>							
2020	69.1	45.9	-18.0	-0.78	-0.97	– 7,556 / 4,797	
2021	70.2	40.3	-11.3	-0.45	-0.36	– 7,556 / 4,797	
2022	66.3	25.1	+0.0	+0.00	+0.00	– 7,556 / 4,797	
2023	40.1	21.4	-22.5	-2.00	-2.39	– 7,556 / 4,797	
2024	19.3	17.6	-39.5	-2.78	-3.64	– 7,556 / 4,797	
2025	17.5	15.9	-39.6	-2.81	-3.61	– 7,556 / 4,797	
2023, ANAF flag	39.7	6.7	–	–	–	– 7,556 / 4,797	
2024, ANAF flag	6.0	0.8	–	–	–	– 7,556 / 4,797	

Micro share = share of the band inferred from the statements to be on the revenue tax that year; the rows marked ANAF flag give instead the share registered for the micro-enterprise tax in the ANAF registry snapshots of 31 August 2023 and June 2024. First stage = change in the treated band’s micro share minus the control band’s, relative to the base year. Per switch = the mean-margin DiD divided by the first stage (a Wald ratio), reported only for the entry design, where the inference is clean in the base year. Bands on two-year average revenue: treated $(0.8, 1.0] \times \text{€}1\text{M}$ throughout. First-stage and per-switch figures are computed from unrounded shares and will not reproduce exactly from the one-decimal shares displayed.

a firm is registered for, puts the treated band at 40 percent micro in August 2023 and 6 percent in June 2024, against 12 and 2 percent for the control band, so by 2024 both bands were almost entirely on profit tax and the treated band had lost the regime on a scale the inference understates. We therefore report the exit effect as an intention-to-treat estimate on the band that lost the regime and compute per-switch effects for the entry design only.

6.2 Results

Table 12 reports the difference in differences on the median margin. Raw pre-period differences are within 0.2 points of zero and the contemporaneous placebos within 0.75; net of the placebos the entry design’s pre-period differences are within 0.05 of zero and the exit design’s are -0.3 and -0.4 (standard errors 0.4 and 0.3), so the correction creates no pre-trend, though the exit design starts from a slightly negative one. Entry in 2018 raises the treated band’s median margin by 0.7 points in the first year (standard error 0.2) from a base of 5.3 percent, 0.6 (0.3) net of placebos; it is significant only at that horizon, at 0.3 (0.3) and 0.5 (0.3) after, which the contaminated control explains in part. Exit in 2023 lowers it by 1.0, 1.6 and 2.0 points over three years from a base of 11.5 percent, -0.8 (0.3), -1.5 (0.4) and -1.9 (0.3) net of placebos, and by 2.0 to 2.8 points against

Table 12: Reported margins around regime switches: narrowed bands with contemporaneous placebos

Year	Horizon	DiD, median margin, pp (SE)	Placebo at €1.5M	Placebo at €2M	Corrected, pp (SE)	DiD, mean margin, pp (SE)	Corrected mean, pp (SE)
<i>2018 entry: bands on 2016–2017 average revenue, cut-off €1M; placebos at €1.5M and €2M</i>							
2015	-2	+0.09 (0.20)	+0.25	-0.06	-0.01 (0.28)	+0.72 (0.47)	+0.93 (0.62)
2016	-1	+0.16 (0.20)	+0.46	-0.04	-0.05 (0.27)	+0.35 (0.36)	+0.24 (0.46)
2017	+0	0.00	+0.00	+0.00	0.00	0.00	0.00
2018	+1	+0.71 (0.21)	+0.28	-0.11	+0.62 (0.26)	+0.94 (0.38)	+0.92 (0.53)
2019	+2	+0.51 (0.23)	+0.44	+0.03	+0.27 (0.29)	+0.69 (0.50)	+0.43 (0.63)
2020	+3	+0.48 (0.23)	+0.14	-0.08	+0.45 (0.28)	+0.60 (0.56)	+0.71 (0.69)
<i>2023 exit: bands on 2021–2022 average revenue, cut-off €1M; placebos at €1.5M and €2M</i>							
2020	-2	-0.06 (0.32)	+0.49	+0.04	-0.33 (0.41)	-0.16 (0.41)	-0.48 (0.57)
2021	-1	+0.07 (0.27)	+0.72	+0.21	-0.39 (0.32)	-0.07 (0.33)	-0.66 (0.44)
2022	+0	0.00	+0.00	+0.00	0.00	0.00	0.00
2023	+1	-0.95 (0.26)	+0.09	-0.36	-0.81 (0.33)	-1.36 (0.38)	-1.07 (0.49)
2024	+2	-1.62 (0.33)	-0.07	-0.20	-1.48 (0.38)	-2.51 (0.42)	-2.37 (0.51)
2025	+3	-1.98 (0.27)	-0.30	+0.06	-1.86 (0.34)	-3.04 (0.45)	-2.86 (0.62)

Balanced panels. Treated band: two-year average revenue in $(0.8, 1.0] \times$ the cut-off; control band: $(1.0, 1.2] \times$ the cut-off; DiD = change in the treated band’s median (or mean, winsorised at ± 1) pre-tax margin minus the change in the control band’s, relative to the base year. Placebos: the same construction at cut-offs of €1.5M and €2M in the same base year, where no regime changed; the placebo columns shown are for medians. Corrected = DiD minus the mean of the two placebos. SE from 200 bootstrap draws over firms. Firms: 5,353/3,760 treated/control in 2018, 7,556/4,766 in 2023; placebo designs 2,200–4,800 per band.

the wider control band. The mean margin, winsorised at ± 1 , moves the same way and a little more: entry raises it by 0.9 points (standard error 0.4) in the first year from a base of 6.8 percent, and exit lowers it by 1.4, 2.5 and 3.0 points (0.4 to 0.5) from a base of 15.7 percent, 9 to 19 percent of the base. Per regime switch, the mean-margin effect divided by the first stage in the same year, a Wald ratio, is 3.7 points against the narrow control band and 2.3 against the wider one, a third to a half of the base mean margin. For the exit design we keep the intention-to-treat figure: 1.0 to 2.0 points on medians, 8 to 17 percent of the base margin, and 1.4 to 3.0 on means, on a band whose registered micro share fell from about two thirds to 6 percent. The exit effect builds over three years, which fits arrangements that take time to unwind.

6.3 Fiscal magnitudes

Table 13 puts the regime in lei for 2023, using the ANAF registry to say who was in it. Micro-enterprises paid 3.3 billion lei of revenue tax, 0.9 to 1.0 percent of their revenue in every band below €500,000, which is the rate the law prescribed; 16 percent of their reported pre-tax profit would have been 10.7 billion, a static gap of 7.3 billion lei. Of that gap, 81 percent comes from firms with revenue below €500,000 and 12 percent from the €500,000 to €1,000,000 band where the 2023 bunchers and the reclassified firms sit. Cash receipts bound the accrual from above rather than reconcile it. The Ministry of Finance’s budget execution for 2023 (Ministerul Finanțelor, 2024) reports 5.49 billion lei for the state-budget chapter that carries the micro-enterprise tax together with withholding on non-residents and smaller items, against 4.76 billion in 2022; neither the execution note nor ANAF’s statistical bulletin separates the micro-enterprise line, so the chapter total is an upper bound on cash receipts of the tax, and how much of the gap to the 3.3 billion

Table 13: Fiscal magnitudes in 2023, million lei: micro-enterprise tax paid against 16 percent of reported pre-tax profit, by revenue band

Revenue band (€, prev. rate)	Firms	Micro firms	Micro tax paid	16% of their profit	Gap	Gap share (%)	Profit-tax firms	Profit tax paid
< 60k	420,646	210,267	306	1,187	880	12	196,922	529
60-250k	164,407	143,787	822	3,486	2,664	36	19,880	335
250-500k	55,483	48,765	785	3,217	2,432	33	6,527	176
500k-1M	31,385	21,463	927	1,819	891	12	9,858	471
> 1M	44,797	3,762	502	982	480	6	40,729	20,104

Micro firms are those registered for the micro-enterprise tax in the ANAF registry of 31 August 2023, profit-tax firms those registered for profit tax; firms above €500,000 registered as micro-enterprises are in-year crossers, taxed on revenue until the quarter of crossing and on profit after it, and their tax paid includes both, as are hospitality firms, which faced no ceiling in 2023. Profit at 16% uses reported pre-tax profit floored at zero and is an upper bound, since Section 6 shows reported margins fall by 8 to 17 percent of their level when firms move to profit tax. The gap is the static revenue difference; the share column is each band’s share of the total gap.

accrued in the statements is withholding and how much is coverage cannot be sized from it. The coverage side is the registry filter: the statement total covers the 455,035 firms registered for the micro-enterprise tax at 31 August 2023 that filed accounts, while ANAF counts 495,448 registrations for the tax at 31 December 2023 against 912,875 a year earlier, registrations rather than filers, so firms that entered or left the regime during the year, including those the 2023 reform pushed out, are outside the total. The registry matters here: the accounts-based inference would put the micro tax at 8.0 billion, because it classifies 10,000 firms above €1,000,000 as micro-enterprises, large profit-tax firms whose credits and exemptions put their tax at 1 to 3 percent of revenue, and the registry puts 40,000 of the 45,000 firms in that band on profit tax. The gap is an upper bound, and the previous subsection says by how much: firms moved to profit tax in 2023 reported margins 8 to 17 percent lower than their base, so a static simulation that applies 16 percent to the profit those firms reported as micro-enterprises overstates the revenue from reclassifying them by that share. Ban and Buciu (2023) simulate revenue across ceilings from zero to €1,000,000 under exactly that assumption; the correction is first order for the reclassified band and second order for the regime as a whole, because the intramarginal firms, not the threshold, carry the cost.

7 The employee condition and other margins

7.1 The employee condition

From 1 January 2023 a micro-enterprise must have at least one employee, tested at 31 December 2022; the rule was enacted in July 2022. Table 14 shows the share of small firms with exactly one employee rising from 31.9 percent in 2021 to 37.6 in 2022 and 40.7 in 2023. Table 15 follows zero-employee firms: 22.3 percent of them moved to exactly one employee in 2022, against 11.1 to 11.3 percent in the two preceding transitions, or 20.3 percent when the sample is limited to firms present since 2020; among zero-employee

Table 14: Employee counts among firms with revenue below €500,000 (% of firms)

Year	0 employees	1	2	5+	Firms
2019	33.6	28.7	13.9	11.8	555,050
2020	33.4	30.6	13.4	11.2	575,855
2021	35.1	31.9	12.1	10.2	558,724
2022	31.1	37.6	11.9	9.1	616,503
2023	28.7	40.7	12.5	8.3	640,536
2024	30.6	40.8	12.2	7.3	686,605
2025	32.6	40.5	11.8	6.4	655,235

Average annual headcount as reported in the statements.

firms above €1,500,000, untouched by the rule, the rate stayed at 5 to 6 percent. That is about 20,000 additional one-employee firms. Of micro-enterprises still without employees at the end of 2022, 82 percent were registered for profit tax in 2023 and 15 percent had an employee by the end of 2023. The hiring came in the enactment year and the rest left the regime. The aggregate shows the scale. ANAF's count of taxpayers registered for the micro-enterprise tax fell from 912,875 at 31 December 2022 to 495,448 at 31 December 2023 and 368,900 at 31 December 2024 (ANAF, 2024, 2025, Table 2.2): the 2023 reform, the employee rule together with the halved ceiling, the three-firm rule and the consulting cap, took 46 percent of registrants out of the regime in one year, and a further 26 percent had gone by the end of 2024, a fall that mixes the 2024 rules with 2023 leavers whose deregistration was recorded only after the notification deadline of 31 March 2024. The registry counts registrations, not filers, so it includes dormant firms and firms that never filed accounts, which is why it exceeds the 862,564 companies with 2022 accounts in our panel; the two series are not the same population, but the fall is the counterpart of Table 15. The implied tax rates of the 2022 hirers suggest many hired late in the year, but rate inference is noisy for one-employee firms, so we do not lean on the timing. Headcount counts a part-time contract as one, so we read this as compliance at the lowest cost rather than job creation.

7.2 Two further margins, in brief

The same population bunches at the VAT exemption threshold, set in lei, with polynomial excess mass of 2.8 to 6.6 in every year, and when the threshold moved from 300,000 to 395,000 lei on 1 September 2025 the spike relocated within the same year. Its move also

Table 15: Transitions of zero-employee firms (revenue below €500,000) into the next year, conditional on filing in both years (%)

Year t	Firms with 0 employees in t filing in $t + 1$	1 employee in $t + 1$	2+ in $t + 1$	Still 0 in $t + 1$
2019	173,878	11.1	4.7	84.1
2020	165,103	11.3	3.5	85.2
2021	178,783	22.3	3.3	74.4
2022	168,742	14.8	2.9	82.3
2023	169,005	9.4	2.9	87.7
2024	178,415	8.1	2.3	89.6

The one-employee condition was enacted in July 2022 and tested at 31 December 2022. Filing coverage differs across years because some annual files are early uploads, so shares are conditional on a statement being present in $t + 1$.

uncovers the €60,000 boundary between the 1 and 3 percent rates, a two-point notch on the whole base, which draws a spike a fifth the size of the VAT spike. Appendix C gives the tables and figures. Firms in the 5 percent below the ceiling are more likely than equally sized and aged firms just above it to have an administrator who registered another firm in the previous eighteen months, 2.7 against 2.0 percent, or a new firm at their address, 12.2 against 9.2 percent; but the combined revenue of firms sharing an administrator does not bunch at the ceiling in 2024, when the law began to test it that way, so we report splitting as a correlation rather than a mechanism. Appendix D gives the tables.

8 Conclusion

A ceiling that moved six times gives a counterfactual density without a fitted curve and reassigns the same firms between a revenue base and a profit base. The first shows that bunching at the ceiling is real, follows the ceiling and the exchange rate, and amounts to 2 to 8 percent of nearby firms and 0.06 to 0.30 percent of all firms, with no deficit just above the ceiling; firms just below it grow a little less than firms further down, so part of the response is an accumulation below the ceiling, and where the rest of the surplus comes from the data do not settle. The second shows that reported margins rise when firms are moved onto the revenue base and fall by 8 to 17 percent of the margin when they are moved off it. Together they say that both the fiscal cost of the regime and the revenue from narrowing it are smaller than static estimates imply: 81 percent of the cost sits with firms well below the ceiling, and the profit those firms would report under profit tax is lower than the profit they report now.

Three design points follow, limited to the margins we measure. The ceiling is a lever on the intramarginal population, not on behaviour: moving it changes the behaviour of a few thousand firms and the bills of hundreds of thousands. The regime’s advantage is concentrated in high-margin services, and a structure that ties the rate to the margin, the minimum-tax design that Sharma et al. (2025) show dominates threshold systems,

would remove both the notch and the incentive to report margins differently on its two sides. Side conditions that can be met with a single contract produce that contract, as the employee rule did; and side conditions together with a halved ceiling removed 46 percent of registered micro-enterprises in a single year, 2023, a change in the regime’s population larger than anything the ceiling alone has done. What the data do not settle is how much of the margin response is misreporting and how much is income shifting between salary and dividends; that needs personnel costs, which the public accounts do not carry.

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A Legal timeline

The regime’s parameters were changed by Ordinance 8/2013 (mandatory regime, €65,000, 3 percent, from 1 February 2013), Ordinance 102/2013 (consulting cap, 2014), Ordinance 50/2015 of October 2015 (€100,000 and employee-dependent rates, 2016, tested on 2015 revenue), Ordinance 3/2017 (€500,000 from 1 February 2017), Ordinance 79/2017 (€1,000,000 from 2018, consulting cap abolished), Ordinance 25/2018 (capital and employee opt-out from April 2018), Ordinance 16/2022 of 15 July 2022 (€500,000, employee condition, three-firm rule, optional regime with permanent exit, from 2023) as amended by Law 370/2022 of 20 December 2022 (revenue test deferred to 2023 revenue), Law 296/2023 (€60,000 rate boundary and activity list, 2024), Ordinance 115/2023 (one-firm rule and linked-enterprise aggregation, 2024), Ordinance 156/2024 of 31 December 2024 (€250,000 for 2025, €100,000 for 2026, consulting cap abolished), Ordinance 89/2025 (flat 1 percent from 2026) and Ordinance 8/2026 (turnover base from 2026). Exchange-rate provisions: Article 47(1)(c) for the year-end test and Article 52 for the in-year test of Law 227/2015, and before 2016 Article 112⁶ of Law 571/2003 with its implementing norms; in every version the in-year test uses the BNR rate at the close of the previous financial year. VAT thresholds: Article 310 of Law 227/2015 as amended by Law 72/2018 (300,000 lei from 1 April 2018) and Ordinance 22/2025 (395,000 lei from 1 September 2025). A full table with Monitorul Oficial references accompanies the replication files. Policy assessments of the regime are in IMF (2025), Consiliul Fiscal (2024), Ban and Buciu (2023) and Lazăr (2024).

B Data details

Regime inference. The gap between gross and net profit also contains the specific tax on hospitality firms in 2017–2022 and other taxes on the result, net of sponsorship and cash-register credits, so ratios such as 0.8 or 2.4 percent occur and a firm with positive tax and a pre-tax loss is most likely, but not certainly, a micro-enterprise. Of firms registered for profit tax, 8 percent show a micro pattern and half paid no tax, which is what loss-making and dormant profit-tax firms look like; 2.6 percent of 2023 filers are not in the registry. In the registry’s column dictionary, obligation 120 is the micro-enterprise income tax and obligation 100 the profit tax.

Employees. Headcount is reported as zero for a further 5 percent of small firms beyond the blanks. Of firms that go from zero to one employee in 2022, 12 percent paid the no-employee rate for most of that year, which a true average would have rounded to zero; late-year hires are therefore counted. The legal test is met by a full-time contract, by part-time contracts adding up to one, or by a paid mandate contract, none of which the

field distinguishes.

Trade register. The register lists administrators as of the snapshot date, not shareholders and not past administrators; the open data has a representatives table only in snapshots from November 2025 onwards, so administrators at the time of the 2018–2023 observations cannot be recovered. Company names carry no usable marker of sole ownership. Nearly all filers, 96 to 99 percent by year, link to at least one administrator, and the linked share is the same across the groups compared. Four percent of administrator records lack a birth date, so their key is the name alone; results are reported with and without them. Names, birth dates and addresses are used only to build hashed keys and appear in no output or replication file. Net turnover, used for the VAT threshold, is blank for a fifth of micro-entity forms but for under 1 percent of firms near that threshold.

C The VAT threshold and the rate boundary

The VAT threshold moved from 300,000 to 395,000 lei on 1 September 2025. In the 2025 distribution of turnover the spike at 300,000 lei is gone and a spike at 395,000 has replaced it, with polynomial excess mass of 3.0 against 0.6 at that location in 2024 (Table 16); part of the spike is mechanical, since the ordinance let firms that crossed 300,000 lei in August 2025 stay unregistered up to the new threshold. The difference-in-bunching estimator is not available at this threshold because a VAT notch exists in every year within 25 percent of the same-quantile location. The polynomial estimates in Table 16 and Figure 5 show excess mass between 2.8 and 6.6 in every year from 2014 to 2024 at bin widths of 0.7 to 0.9 percent of the threshold, against 0.4 to 2.0 for the micro-enterprise notch at bin widths of 1.5 to 2 percent. The legal base is the turnover of Article 310(2), which differs from net turnover by exempt and ancillary items. In 2024 the €60,000 rate boundary and the VAT threshold are 1,500 lei apart and the combined spike has polynomial excess mass of 4.9; in 2025, with the VAT threshold at 395,000 lei, a spike remains at 296,000 to 298,000 lei, the lei value of €60,000 at the end-2024 rate, with excess mass of 0.9 (Table 17, Figure 4). The residual spike is where the rate boundary predicts it. Among micro-enterprises outside the listed activities the 296,000 to 298,000 lei bin holds 67 percent more firms than the neighbouring bins and the 298,000 to 300,000 bin 40 percent more; among micro-enterprises in the listed activities, taxed at 3 percent regardless of revenue, and among profit-tax firms, neither bin is elevated (Table 18). Firms subject to the rate boundary respond to it and only they do, but the response is a fifth the size of the VAT response; the activity list already puts software, legal and medical firms on 3 percent regardless of revenue, and those are the high-margin firms for which two points of revenue matter most.

Table 16: Bunching of net turnover at the VAT registration threshold

Year	VAT threshold (lei)	Firms in window	b (polynomial)
2014	220,000	83,751	2.84
2015	220,000	91,256	4.21
2016	220,000	97,622	4.76
2017	220,000	105,826	5.39
2018	300,000	102,168	3.64
2019	300,000	113,716	5.09
2020	300,000	111,913	4.90
2021	300,000	117,918	5.78
2022	300,000	135,830	6.61
2023	300,000	145,471	6.62
2024	300,000	157,977	6.09
2025	300,000	157,676	0.79
2025	395,000	147,085	3.03

Polynomial counterfactual (degree 5, three bins excluded on each side), 2,000-lei bins, window $\pm 40\%$. The threshold rose to 395,000 lei on 1 September 2025; the two 2025 rows test the old and the new location (at 395,000 lei the 2024 placebo gives 0.62).

Table 17: The 1%/3% rate boundary at €60,000 and the VAT threshold, 2023–2025

Year	€60,000 in lei	VAT threshold (lei)	b at €60,000	b at 300,000	Modal 2,000-lei bin, 250–450k	Firms in modal / median bin
2023	296,844	300,000	3.17	5.89	298,000–300,000	5,215 / 742
2024	298,476	300,000	4.88	5.53	298,000–300,000	5,121 / 801
2025	298,446	300,000 (to Aug. 2025)	0.94	0.92	296,000–298,000	1,910 / 1,101

Polynomial excess mass, 1,500-lei bins, window $\pm 30\%$. In 2023 no rate boundary existed; in 2024 the boundary and the VAT threshold lie 1,500 lei apart; in 2025 the VAT threshold moved to 395,000 lei on 1 September.

Table 18: The 296–300k lei spike in 2025 revenue by tax regime and listed activity

Group in 2025	Firms, 280–320k lei	296–298k	298–300k	Reference per bin	Excess 296–298k (%)	Excess 298–300k (%)
micro, not listed	15,247	1,301	1,089	780	+67	+40
micro, listed activity	1,991	102	87	107	-5	-19
profit tax	3,479	185	176	182	+2	-3
other	8,274	322	455	401	-20	+13

2,000-lei bins. Reference = mean firms per bin over 288–296k and 300–308k. Micro = inferred from the statements to be on the revenue tax; listed activity = CAEN codes taxed at 3% regardless of revenue (software, hospitality, legal, medical). The €60,000 boundary at the end-2024 rate is 298,446 lei; the VAT threshold was 300,000 lei until 31 August 2025.

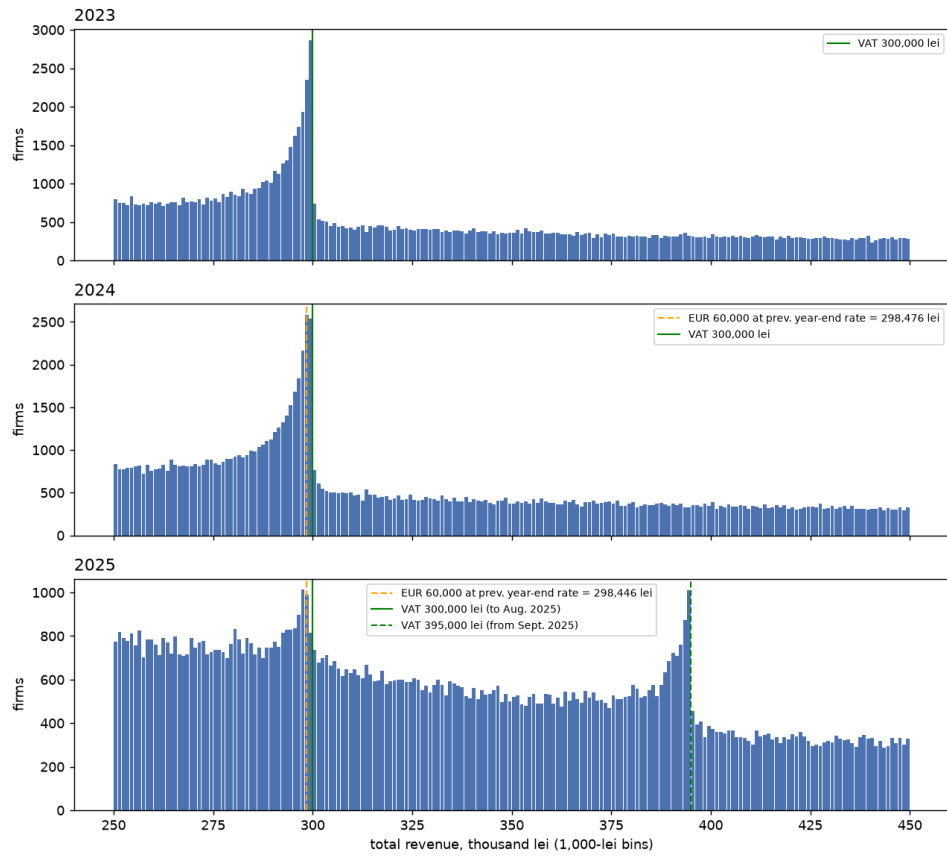


Figure 4: Total revenue from 250,000 to 450,000 lei, 2023–2025, 1,000-lei bins, with the €60,000 rate boundary and the VAT thresholds.

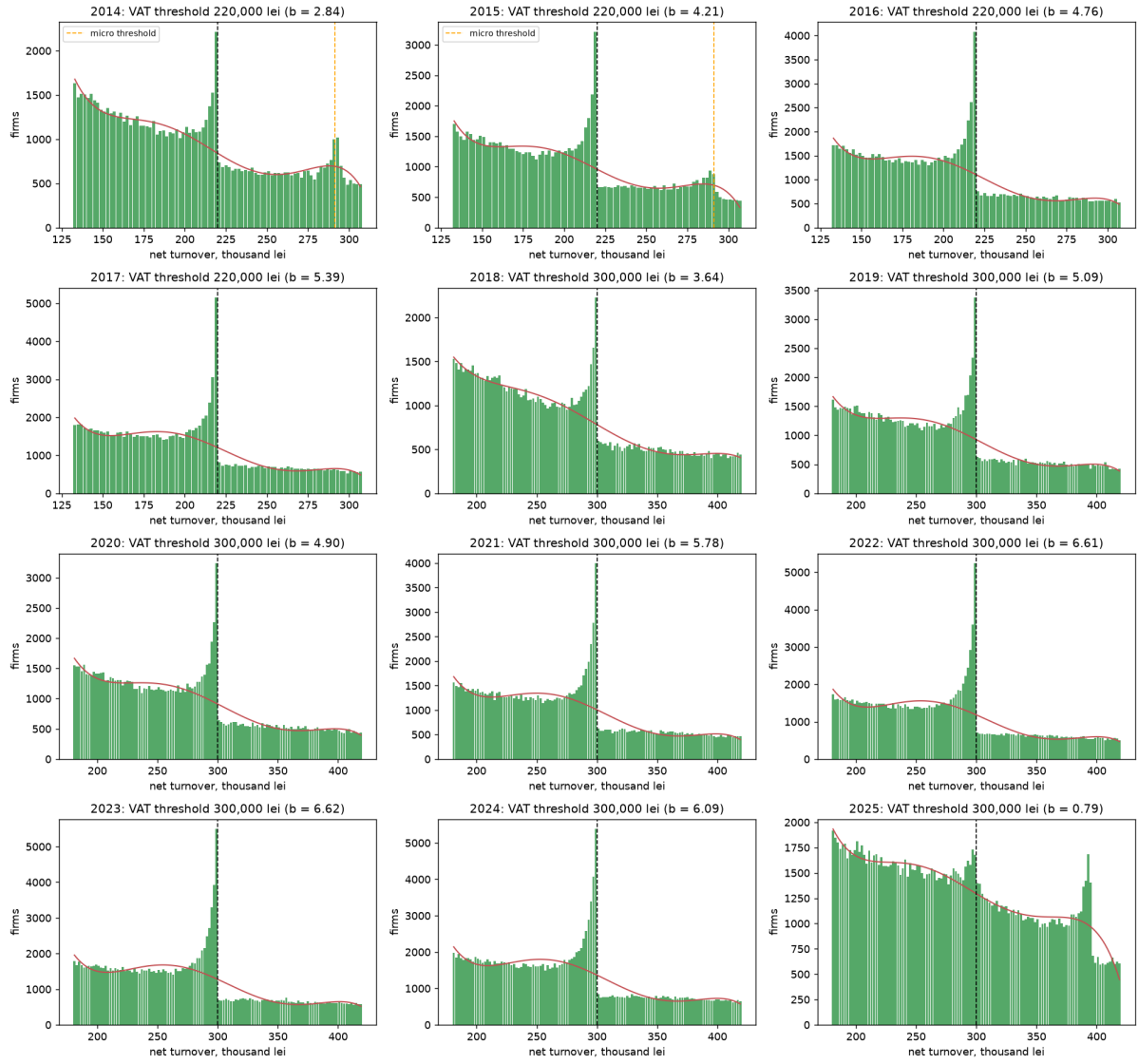


Figure 5: Net turnover around the VAT registration threshold, 2,000-lei bins, with a polynomial counterfactual.

Table 19: Sibling-firm indicators by position relative to the threshold, SRLs, pooled 2018–2023

Measure	Bunchers	Just above	10–25% below	25–100% above	z
Administrator registered another firm, prior 18 months	2.71	1.98	1.72	1.91	4.3
New firm at the same address, prior 18 months	12.23	9.18	8.72	8.37	8.6
Administrator runs another firm, any date	17.09	15.88	15.16	17.46	2.6

Pooled shares, %. z tests bunchers against firms just above the threshold with standard errors clustered by firm. At the small ceilings of 2015–2017 the corresponding shares for bunchers and firms just above are 1.59 versus 1.37 ($z = 1.5$) on the administrator measure, 7.17 versus 5.34 ($z = 6.3$) on the address measure and 12.42 versus 11.38 ($z = 2.7$) on the any-date measure. Standardising the just-above group to the bunchers’ age distribution gives 2.70 versus 2.20 on the administrator measure and 12.26 versus 10.23 on the address measure (Appendix Table 20). Administrators as recorded in the ONRC snapshot of 2 September 2026; persons administering more than 50 firms and addresses hosting more than 20 firms excluded. Bunchers: 0–5% below the threshold; just above: 0–10% above.

D Splitting

Table 19 compares limited-liability companies in the 5 percent below the ceiling with firms just above it, which are the same size, and with firms further below and well above, pooled over 2018–2023 with standard errors clustered by firm. Dropping administrators without a birth date gives 2.4 against 1.7 percent, the same ratio. The share whose administrator runs any other firm, regardless of date, differs little between bunchers and firms just above ($z = 2.6$), so the timing measures carry the information. Bunchers are younger than firms just above, 13 percent under three years old against 8, and Table 20 compares within age bands: bunchers exceed the just-above group in every band, and standardised to the bunchers’ age distribution the premium is 23 percent on the administrator measure and 20 percent on the address measure. Two checks follow from the 2024 rules, which limit a person to one micro-enterprise and test the ceiling on linked enterprises combined. The administrator premium disappears in 2024, 1.4 percent for both groups against 2.0 and 1.6 in 2023, while a gap remains on the address measure; but Table 21 shows no excess mass in the combined revenue of firms sharing an administrator at €500,000 in 2024, and firms inside such groups bunch individually at the same rate as single firms. Either the aggregation rule was not enforced or noticed in its first year, or common administration is too weak a proxy for the ownership links the rule is written on. Firms that lost the regime when the ceiling fell registered new firms at the same rate, relative to the band below, as placebo cohorts.

Table 20: Sibling-firm indicators by firm age, bunchers versus firms just above the threshold, pooled 2018–2023 (%)

Firm age (years)	Bunchers, n	Just above, n	Admin. new firm: bunchers	just above	Same-address new firm: bunchers	just above
0-2	2,166	1,482	6.97	5.47	26.13	20.51
3-5	2,544	2,332	3.26	2.57	18.20	16.51
6-10	3,662	3,826	2.48	2.12	13.68	11.53
11+	8,158	10,187	1.50	1.25	6.09	4.96

Age = year minus year of registration in the trade register. Standardising the just-above group to the bunchers' age distribution gives 2.70 versus 2.20 (administrator measure, +23%) and 12.26 versus 10.23 (address measure, +20%).

Table 21: Bunching of combined revenue of firms sharing an administrator, and firm-level bunching inside groups

Year	Groups (2–5 SRLs, one administrator)	Groups in window	b , combined revenue at €500k	Excess groups	Firms in groups: % in 5% below own notch	Single firms: same
2021	13,563	970	0.11	3	19.5	18.1
2022	15,156	1,118	0.58	16	18.9	17.5
2023	15,860	1,304	0.90	29	21.5	20.1
2024	17,323	1,417	0.07	3	16.2	16.9
2025	18,044	1,335	0.62	20	15.8	14.5

From 2024 the ceiling is tested on the combined revenue of linked enterprises. Combined revenue of each administrator's firms, window $\pm 20\%$ around €500,000 at the previous year-end rate, controls = same-quantile density of combined revenue in 2019–2022 excluding years where the individual €1M notch is within 25%. Last two columns: share of firms within 20% of that year's individual notch that sit in the 5% just below it.

E Additional tables

Table 22 lists the admissible control years for each notch with the matched revenue and distance to the nearest notch. Table 23 gives further robustness checks. One of its entries records the admissibility rule at work: with 2015 admitted as a control for the 2016 notch, before the €100,000 ceiling for 2016 was on the notch list, the pre-years-only estimate for 2016 was 1.30; with 2015 excluded it is 1.69, in line with the other columns. Three other checks are not tabulated: the excess counts vary across the 3, 5 and 8 percent region definitions by 6 to 50 percent, mostly at the two extra notches; dropping hospitality firms, which faced no ceiling in 2023, raises the 2023 estimate from 2.99 to 3.14; newly created firms make up 11 percent of the 2023 excess, against 7 percent in 2021 and 13 percent in 2024. Table 24 gives the origin of firms just below €500,000 in the notch year and in placebo years. Table 25 gives the wide-band event study. Table 26 gives the reduced-form elasticities discussed in Section 4. Table 27 extends the count-based comparison to a window of ± 50 percent, Table 28 compares the growth of firms below the notch in notch and control years, Table 29 gives margins around the 2015 year-end notch, and Table 30 follows firms on either side of it into the next year.

Table 22: Admissible control years, matched locations and distance to the nearest notch

Notch	Control year: matched revenue L_c^* (distance to nearest notch in that year)
2014: in-year, €65,000	2016: 0.32M (32%); 2017: 0.33M (34%); 2022: 0.44M (31%); 2023: 0.46M (35%); 2024: 0.47M (36%)
2015: in-year, €65,000	2016: 0.30M (26%); 2017: 0.30M (27%); 2023: 0.43M (30%); 2024: 0.43M (31%)
2016: in-year, €100,000	2014: 0.42M (30%); 2017: 0.48M (54%); 2018: 0.50M (40%); 2019: 0.52M (42%); 2020: 0.49M (39%); 2021: 0.57M (47%); 2022: 0.62M (52%); 2023: 0.66M (54%); 2024: 0.66M (54%)
2017: in-year, €500,000	2014: 2.09M (86%); 2015: 2.25M (80%); 2016: 2.25M (80%); 2018: 2.42M (88%); 2019: 2.50M (86%); 2020: 2.39M (88%); 2021: 2.80M (74%); 2025: 3.13M (60%)
2018: in-year, €1,000,000	2014: 4.19M (93%); 2015: 4.49M (90%); 2016: 4.44M (90%); 2017: 4.54M (50%); 2023: 5.83M (58%); 2024: 5.88M (58%); 2025: 6.26M (80%)
2019: in-year, €1,000,000	2014: 4.12M (93%); 2015: 4.41M (90%); 2016: 4.37M (90%); 2017: 4.46M (49%); 2023: 5.73M (57%); 2024: 5.79M (57%); 2025: 6.16M (80%)
2020: in-year, €1,000,000	2014: 4.42M (93%); 2015: 4.73M (90%); 2016: 4.69M (90%); 2017: 4.76M (52%); 2023: 6.15M (60%); 2024: 6.21M (60%); 2025: 6.60M (81%)
2021: in-year, €1,000,000	2014: 3.92M (93%); 2015: 4.19M (89%); 2016: 4.15M (89%); 2017: 4.24M (46%); 2023: 5.44M (54%); 2024: 5.48M (55%); 2025: 5.84M (79%)
2022: in-year, €1,000,000	2014: 3.56M (92%); 2015: 3.82M (88%); 2016: 3.79M (88%); 2017: 3.87M (41%); 2023: 4.94M (50%); 2024: 5.01M (50%); 2025: 5.32M (77%)
2023: in-year, €500,000	2014: 1.79M (84%); 2015: 1.94M (77%); 2016: 1.94M (77%); 2018: 2.09M (86%); 2019: 2.16M (86%); 2020: 2.07M (86%); 2021: 2.42M (88%); 2025: 2.70M (54%)
2024: in-year, €500,000	2014: 1.75M (83%); 2015: 1.91M (76%); 2016: 1.90M (76%); 2018: 2.05M (85%); 2019: 2.12M (86%); 2020: 2.03M (85%); 2021: 2.37M (87%); 2025: 2.65M (53%)
2025: in-year, €250,000	2014: 0.82M (64%); 2015: 0.90M (50%); 2016: 0.89M (49%); 2017: 0.92M (76%); 2018: 0.97M (69%); 2019: 1.01M (70%); 2020: 0.96M (69%); 2021: 1.11M (73%); 2022: 1.21M (75%); 2023: 1.25M (76%); 2024: 1.23M (76%)
2015: year-end, €100,000	2014: 0.40M (28%); 2017: 0.46M (52%); 2018: 0.48M (38%); 2019: 0.50M (40%); 2020: 0.47M (37%); 2021: 0.55M (45%); 2022: 0.60M (50%); 2023: 0.63M (53%); 2024: 0.64M (53%)
2022: announced, €500,000	2014: 1.69M (83%); 2015: 1.84M (75%); 2016: 1.83M (75%); 2018: 1.97M (85%); 2019: 2.05M (85%); 2020: 1.96M (85%); 2021: 2.28M (87%); 2025: 2.56M (51%)
2025: year-end, €100,000	2016: 0.34M (31%); 2017: 0.35M (37%); 2021: 0.42M (29%); 2022: 0.46M (35%); 2023: 0.49M (39%); 2024: 0.50M (40%)

L_c^* = revenue at the notch's quantile in the control year. Distance = $|n/L_c^* - 1|$ for the nearest known notch n of that year (micro ceiling, future ceiling, €60,000 boundary or VAT threshold); admissible if above 25%. The closest cases are 26–34%.

Table 23: Further robustness checks of the difference-in-bunching estimates

Notch	b	Quantile-corrected	Pre-years only	Post-years only	Continuing firms	n continuing	Jackknife SE
2014: in-year, €65,000	1.15	1.14	–	1.15	1.15	31,696	0.04
2015: in-year, €65,000	1.71	1.70	–	1.71	1.71	30,382	0.02
2016: in-year, €100,000	1.64	1.63	1.69	1.63	1.52	29,322	0.04
2017: in-year, €500,000	0.76	0.77	0.62	0.85	0.72	16,321	0.05
2018: in-year, €1,000,000	1.41	1.36	1.44	1.37	1.35	11,010	0.04
2019: in-year, €1,000,000	1.67	1.75	1.64	1.70	1.58	12,513	0.06
2020: in-year, €1,000,000	1.75	1.87	1.64	1.90	1.67	11,535	0.09
2021: in-year, €1,000,000	2.15	2.19	2.23	2.04	2.08	13,668	0.08
2022: in-year, €1,000,000	1.92	1.97	1.82	2.05	1.77	14,756	0.09
2023: in-year, €500,000	2.99	2.99	3.02	2.82	2.80	26,100	0.05
2024: in-year, €500,000	1.62	1.65	1.64	1.52	1.58	24,967	0.04
2025: in-year, €250,000	0.73	0.72	0.73	–	0.61	35,470	0.04
2015: year-end, €100,000	-0.03	-0.04	0.09	-0.05	-0.02	26,098	0.04
2022: announced, €500,000	0.80	0.79	0.80	0.83	0.83	21,379	0.03
2025: year-end, €100,000	1.47	1.47	1.47	–	1.43	51,879	0.04

Quantile-corrected: control quantile shifted down by B/N_t . Pre/post: only control years before/after the treated year. Continuing: firms with positive revenue in $t - 1$, t and $t + 1$. Jackknife SE over control years.

Table 24: Origin of firms just below €500,000: share that was above €500,000 the year before

Year	Status	Firms 0–6% below	% above €500k in $t - 1$	% in 120–200% band in $t - 1$	Firms 14–20% below	% above in $t - 1$	Ratio
2019	placebo	2,999	23.6	8.3	3,922	12.3	1.92
2020	placebo	2,895	40.7	17.5	3,817	25.0	1.63
2021	placebo	3,058	19.4	7.3	3,999	9.9	1.96
2023	notch year	6,646	29.6	15.1	5,166	16.6	1.78
2024	post-transition	5,659	21.7	7.3	5,403	14.0	1.55

Positions relative to €500,000 at the previous year-end rate in every year, whether or not it was a threshold. Ratio = bunchers' share from above divided by the comparison group's share.

Table 25: Reported pre-tax margins around regime switches induced by ceiling changes

Year	Micro share T / C (%)	Median margin T (%)	Median margin C (%)	DiD median (pp)	DiD mean (pp)	Median revenue T / C (k€)
<i>2018 increase: (500k, 1M] vs (1M, 2M] on 2017 revenue</i>						
2015	15 / 14	5.39	5.07	-0.32	-0.39	559 / 1,108
2016	14 / 14	5.80	5.41	-0.25	-0.43	628 / 1,232
2017	27 / 15	6.00	5.36	+0.00	+0.00	684 / 1,351
2018	72 / 18	6.37	4.93	+0.80	+0.85	728 / 1,438
2019	61 / 22	6.22	4.89	+0.69	+0.65	746 / 1,466
2020	58 / 26	6.07	5.06	+0.37	-0.08	675 / 1,342
<i>2023 cut: (500k, 1M] vs (1M, 2M] on 2022 revenue</i>						
2020	74 / 51	9.84	7.70	-0.70	-1.14	462 / 912
2021	75 / 48	11.87	9.36	-0.33	-0.51	576 / 1,090
2022	75 / 36	11.76	8.92	+0.00	+0.00	696 / 1,351
2023	55 / 24	8.40	6.69	-1.13	-1.49	676 / 1,358
2024	24 / 18	6.43	5.69	-2.10	-2.49	691 / 1,381
2025	18 / 17	5.80	5.35	-2.39	-3.22	686 / 1,366
<i>Placebo: same bands on 2019 revenue, no regime change</i>						
2017	46 / 24	7.35	5.66	-1.47	-1.60	533 / 1,099
2018	74 / 37	8.24	5.82	-0.74	-0.88	621 / 1,249
2019	71 / 26	8.95	5.79	+0.00	+0.00	692 / 1,364
2020	70 / 21	7.85	5.54	-0.85	-0.98	640 / 1,270
2021	66 / 29	9.43	6.63	-0.36	-1.17	718 / 1,421

Balanced panels of firms with positive revenue in every year shown. T = firms whose base-year revenue placed them in the band that changed regime; C = firms in the band above that did not. DiD = change in T minus change in C relative to the base year (2017, 2022, 2019). Margins are pre-tax profit over revenue, winsorised at ± 1 . The placebo shows the drift induced by selecting on base-year revenue alone.

Table 26: Reduced-form elasticities under the Kleven–Waseem approximation, reported for completeness

Notch	τ	median buncher margin	Δt (pp)	$\Delta z^*/z^*$ from b (%)	e from b	hole width (%)	e from hole
2014: in-year, €65,000	3%	0.160	-0.44	1.15	undefined (dt<=0)	–	undefined (dt<=0)
2015: in-year, €65,000	3%	0.169	-0.30	1.71	undefined (dt<=0)	20	undefined (dt<=0)
2016: in-year, €100,000	1%	0.134	+1.14	1.64	0.012	20	1.73
2017: in-year, €500,000	1%	0.089	+0.42	0.76	0.007	20	4.67
2018: in-year, €1,000,000	1%	0.106	+0.70	1.41	0.014	18	2.3
2019: in-year, €1,000,000	1%	0.117	+0.87	1.67	0.016	18	1.84
2020: in-year, €1,000,000	1%	0.132	+1.11	1.75	0.014	19	1.61
2021: in-year, €1,000,000	1%	0.160	+1.56	2.15	0.015	19	1.15
2022: in-year, €1,000,000	1%	0.166	+1.66	1.92	0.011	20	1.2
2023: in-year, €500,000	1%	0.200	+2.20	2.99	0.02	–	hole not identified
2024: in-year, €500,000	3%	0.151	-0.58	1.62	undefined (dt<=0)	19	undefined (dt<=0)
2025: in-year, €250,000	3%	0.135	-0.84	0.73	undefined (dt<=0)	20	undefined (dt<=0)
2015: year-end, €100,000	3%	0.055	-2.12	-0.03	undefined (dt<=0)	–	undefined (dt<=0)
2022: announced, €500,000	1%	0.140	+1.24	0.80	0.003	18	1.29
2025: year-end, €100,000	3%	0.234	+0.74	1.47	0.014	20	2.61

$\Delta t = 0.16 \times \text{margin} - \tau$ is the change in the average tax rate on revenue at the notch for a firm with the median reported margin of bunchers; $e = (\Delta z^*/z^*)^2 / (2\Delta t / (1 - \tau))$. Two proxies for $\Delta z^*/z^*$: b with 1% bins, and the width above the notch at which the cumulative deficit offsets the excess (the hole), which is not reached before the window edge in most years. The proxies differ by three orders of magnitude and $\Delta t \leq 0$ in the 3% years, which is why the text does not rely on these numbers.

Table 27: Count-based differences by distance from the notch, window $\pm 50\%$

Notch	-50 to -20%	-20 to -5%	-5 to 0%	0 to 10%	10 to 20%	20 to 30%	30 to 40%	40 to 50%
2014: in-year, €65,000	-10,248 (-17)	-620 (-4)	+1,043 (+25)	+425 (+6)	-384 (-6)	-435 (-7)	-782 (-13)	+37 (+1)
2015: in-year, €65,000	-7,227 (-11)	+120 (+1)	+1,554 (+35)	-778 (-10)	-668 (-9)	-567 (-9)	-446 (-8)	-663 (-12)
2016: in-year, €100,000	-6,466 (-14)	+692 (+5)	+1,557 (+39)	-296 (-4)	-470 (-7)	-231 (-4)	-62 (-1)	-167 (-4)
2017: in-year, €500,000	+447 (+2)	+245 (+3)	+399 (+19)	-54 (-1)	-65 (-2)	-41 (-2)	-30 (-1)	-95 (-4)
2018: in-year, €1,000,000	+896 (+5)	+300 (+6)	+477 (+35)	-139 (-6)	-74 (-4)	-18 (-1)	+13 (+1)	+0 (-0)
2019: in-year, €1,000,000	+1,091 (+6)	+721 (+13)	+736 (+49)	+107 (+4)	-173 (-8)	-61 (-3)	-38 (-2)	+33 (+2)
2020: in-year, €1,000,000	+1,158 (+6)	+627 (+12)	+739 (+50)	+16 (+1)	-121 (-6)	-38 (-2)	+4 (+0)	-38 (-3)
2021: in-year, €1,000,000	+1,467 (+8)	+1,007 (+18)	+1,057 (+67)	+130 (+5)	-132 (-6)	-23 (-1)	-8 (-0)	-100 (-6)
2022: in-year, €1,000,000	+404 (+2)	+861 (+13)	+1,049 (+56)	+7 (+0)	-36 (-1)	-21 (-1)	+47 (+2)	-1 (-0)
2023: in-year, €500,000	+2,661 (+8)	+2,064 (+19)	+2,749 (+92)	+220 (+4)	-309 (-7)	-209 (-5)	-74 (-2)	-158 (-5)
2024: in-year, €500,000	+1,469 (+4)	+1,424 (+12)	+1,551 (+48)	+73 (+1)	-115 (-2)	-41 (-1)	-127 (-3)	-111 (-3)
2025: in-year, €250,000	+1,795 (+4)	+762 (+5)	+907 (+20)	+155 (+2)	-116 (-2)	-28 (-0)	-76 (-1)	+4 (+0)
2015: year-end, €100,000	-6,156 (-13)	-919 (-7)	-155 (-4)	+235 (+3)	-183 (-3)	-24 (-0)	+125 (+2)	+30 (+1)
2022: announced, €500,000	-432 (-1)	+399 (+4)	+580 (+19)	-58 (-1)	-245 (-5)	-79 (-2)	+14 (+0)	-49 (-2)
2025: year-end, €100,000	+7,705 (+9)	+2,751 (+13)	+2,850 (+46)	+1,116 (+10)	-467 (-5)	-481 (-5)	-12 (-0)	+317 (+4)

Treated minus same-quantile counterfactual firms in each slice, with the difference as a percent of the counterfactual firms in the slice in parentheses. Counterfactual scaled to the year’s filers; control years as in the baseline, some of which have another notch inside the wider window.

Table 28: Revenue growth from t to $t + 1$ of firms below the ceiling in notch years and in control years

Year	Median growth, 5–20% below: notch	controls	Share crossing next year’s notch: notch	controls	Growth gap, 5–20% minus 20–35% below: notch	controls	Median growth, 0–5% below: notch	controls
2014	+4.2	+2.5	37.5	43.0	+1.6	+2.5	-0.1	+3.1
2018	+5.2	+3.8	30.8	31.1	-0.5	-0.1	+2.3	+3.3
2019	-5.3	+3.9	20.2	20.7	-0.4	-0.1	-4.3	+3.0
2020	+10.3	+3.7	38.9	47.8	-2.7	-0.1	+4.4	+4.0
2021	+10.3	+3.8	39.6	42.8	-1.4	-0.2	+5.2	+4.2
2023	+1.1	+4.8	28.3	33.8	-0.5	-0.2	-0.2	+5.0

Years in which the ceiling was unchanged in $t + 1$. Bands are defined relative to the notch location in t ; in control years, relative to the same quantile. Growth = revenue in $t + 1$ over revenue in t , minus one, in percent, median over firms with positive revenue in both years. Crossing = revenue in $t + 1$ above the in-year notch of $t + 1$ (in control years, above the same quantile of $t + 1$). Controls: mean over admissible control years whose following year is also notch-free at the matched location.

Table 29: Margins around the €100,000 ceiling for 2016 status, tested on 2015 revenue

Year	Band	Firms	Median margin	Margin < 6.25% (%)	Margin < 18.75% (%)
2014	5-0% below	3,575	0.025	63	82
2014	0-3% above	2,131	0.024	65	84
2014	3-10% above	4,705	0.027	63	82
2014	10-20% above	5,945	0.024	64	84
2015	5-0% below	3,720	0.055	52	76
2015	0-3% above	2,614	0.038	60	83
2015	3-10% above	4,658	0.041	58	81
2015	10-20% above	5,999	0.045	56	80
2016	5-0% below	4,832	0.108	41	61
2016	0-3% above	3,406	0.137	36	57
2016	3-10% above	4,693	0.049	54	77
2016	10-20% above	5,945	0.049	54	78

Bands relative to €100,000 at the end-2015 rate, 452,450 lei, in 2015 and to the same quantile of the 2014 and 2016 distributions. In 2016 the location coincides with that year's in-year notch.

Table 30: Firms on either side of the €100,000 ceiling for 2016 status, followed into the next year

Cohort	Band in t	Firms in $t+1$	Micro pattern in $t+1$ (%)	Profit-tax pattern (%)	No tax (%)	Median tax / revenue, $t+1$ (%)	Median revenue growth (%)	Below the location in $t+1$ (%)
2014→2015	5-0% below	3,396	15	22	36	0.49	+8.4	43
2014→2015	0-3% above	2,002	15	21	37	0.39	+9.3	38
2014→2015	3-10% above	4,457	14	24	35	0.47	+7.7	33
2015→2016	5-0% below	3,531	57	9	13	1.00	+1.3	53
2015→2016	0-3% above	2,489	21	21	32	0.54	+4.2	42
2015→2016	3-10% above	4,416	19	22	31	0.62	+2.5	39
2017→2018	5-0% below	4,186	89	0	2	1.00	+5.3	46
2017→2018	0-3% above	2,363	89	0	2	1.00	+6.4	39
2017→2018	3-10% above	4,933	89	0	2	1.00	+5.7	35

Bands relative to €100,000 at the end-2015 rate, 452,450 lei, on 2015 revenue; the 2014 and 2017 cohorts take the same quantile of their year's distribution, where no ceiling applied. Regime patterns inferred from the statements of $t+1$. Growth = revenue in $t+1$ over revenue in t , minus one; below the location = share whose $t+1$ revenue is below the band's reference value.

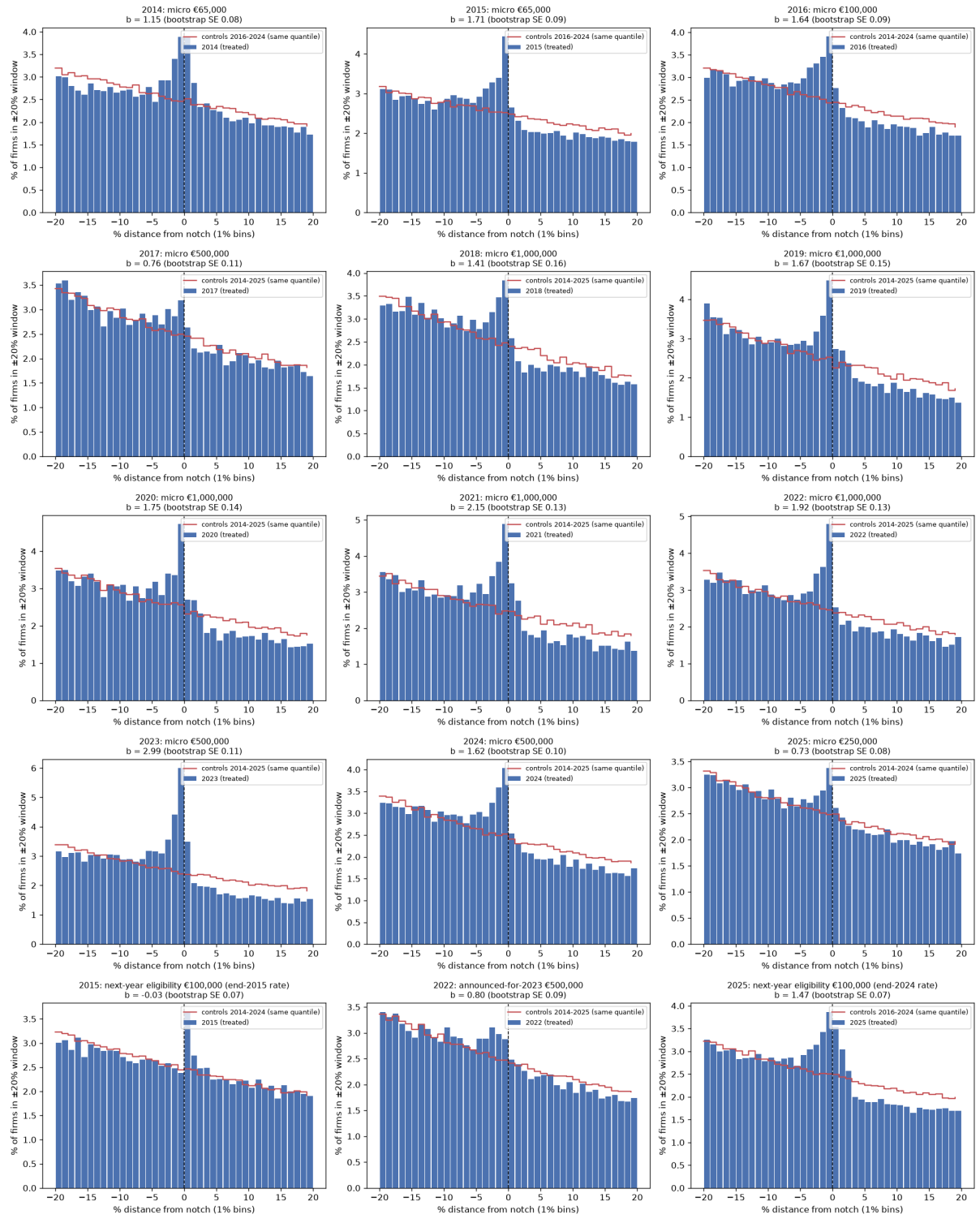


Figure 6: Treated-year density (bars) and same-quantile control density (line) around each of the fifteen notches, bins of 1 percent of the notch location.

F Additional figures

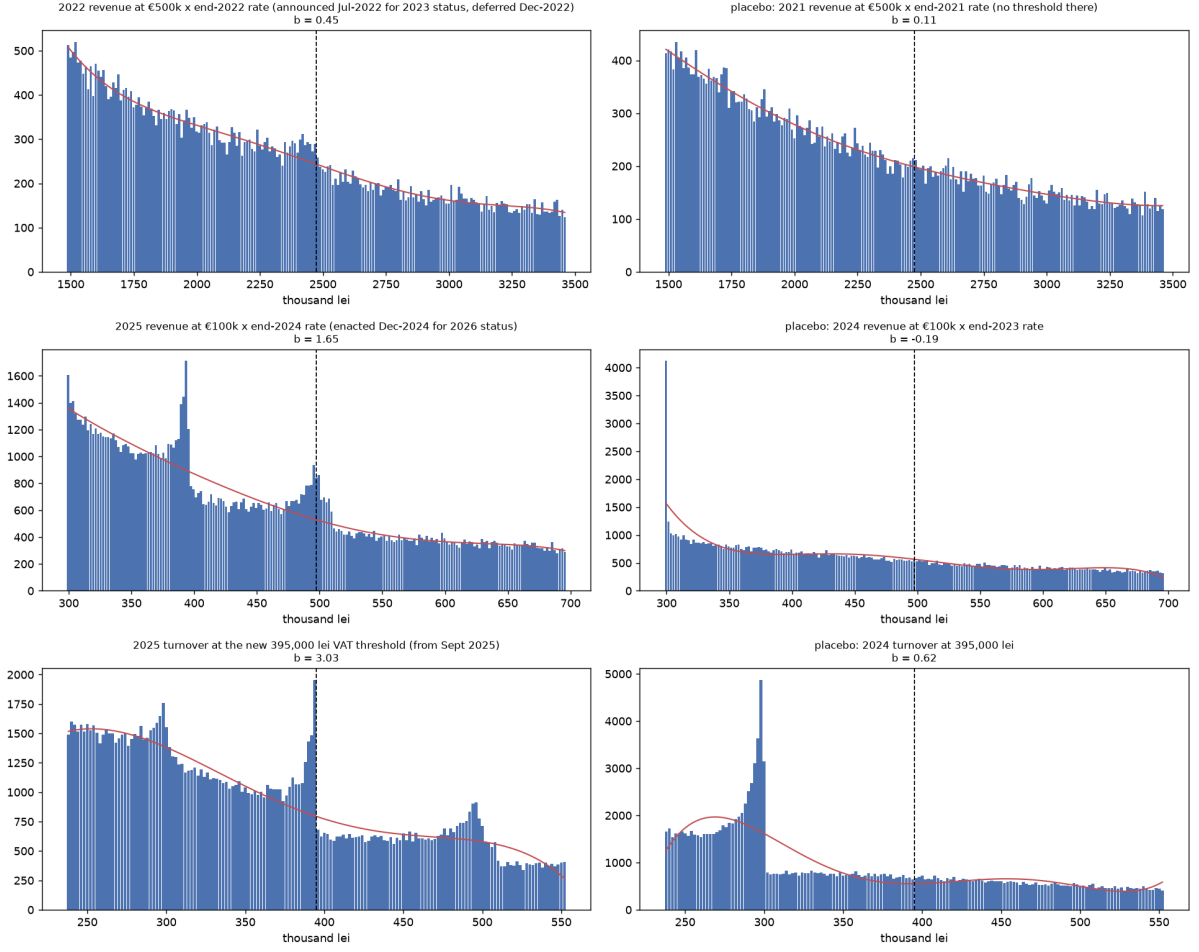


Figure 7: Revenue around announced or enacted future ceilings and the new VAT threshold, with placebo years. Top: 2022 revenue at €500,000, announced in July 2022 for 2023 status and deferred in December 2022, with 2021 placebo. Middle: 2025 revenue at €100,000 at the end-2024 rate, enacted December 2024 for 2026 status, with 2024 placebo. Bottom: 2025 turnover at 395,000 lei with 2024 placebo. Polynomial counterfactuals for reference.

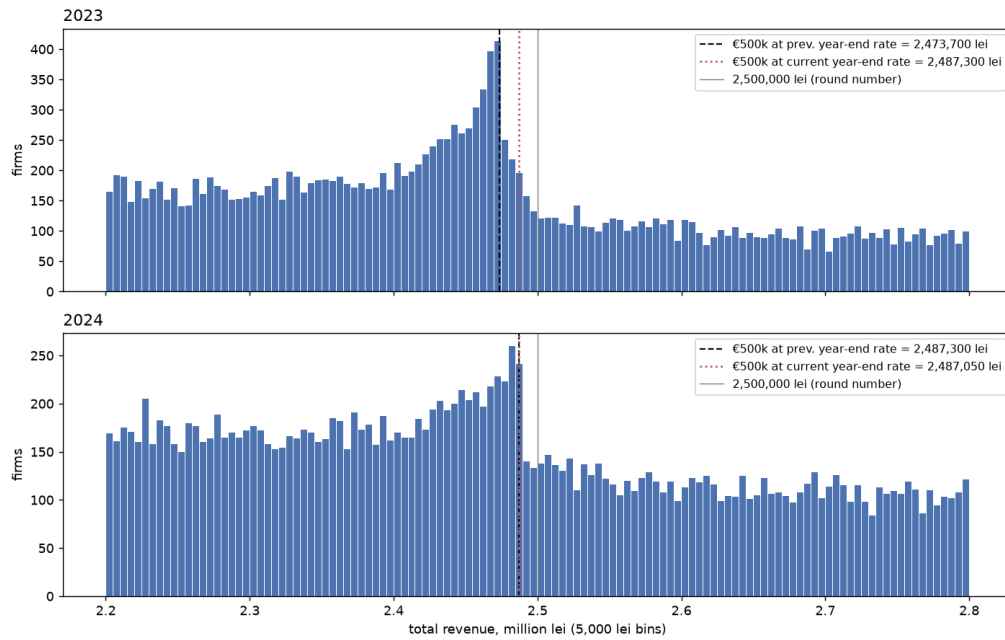


Figure 8: Total revenue in lei around €500,000, 2023 and 2024, 5,000-lei bins. The grey line marks 2,500,000 lei, the nearest round number.

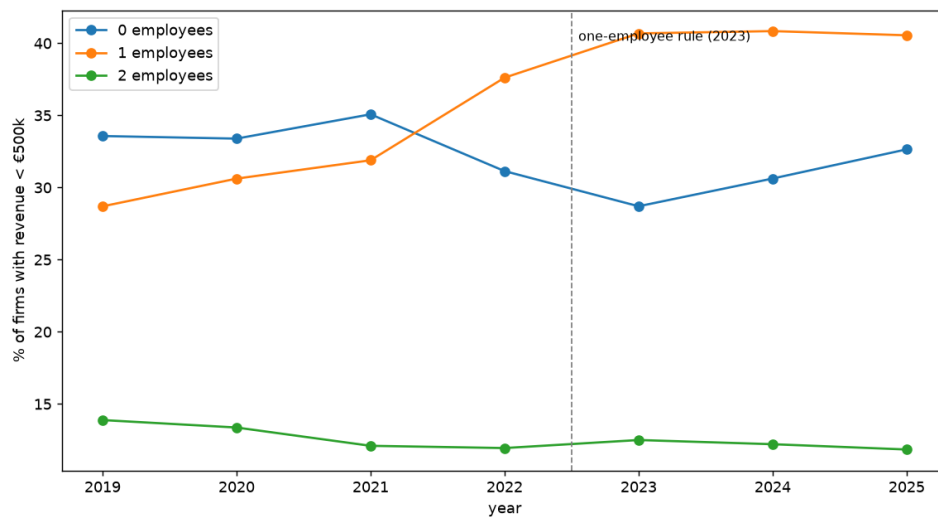


Figure 9: Share of firms with revenue below €500,000 reporting zero, one and two employees, 2019–2025.